Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 1 of 33 · Question 8 part b · 5 marks

2022 JAN · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

8 (a)	$90\pi = \pi r^2 h \Rightarrow h = \frac{90}{r^2}$ $S = 2\pi r^2 + 2\pi rh \Rightarrow S = 2\pi r^2 + 2\pi r \times \frac{90}{r^2}$ $S = 2\pi r^2 + \frac{2\times 90\pi}{r} = 2\pi r^2 + \frac{180\pi}{r}$ *	B1 M1 A1cso [3]
(b)	$\frac{dS}{dr} = 4\pi r - \frac{180\pi}{r^2}$ $\frac{dS}{dr} = 0 \Rightarrow 4\pi r - \frac{180\pi}{r^2} = 0 \Rightarrow 4\pi r = \frac{180\pi}{r^2} \Rightarrow r^3 = 45 \Rightarrow r = ...$ $r = 3.55689... \Rightarrow r \approx 3.56$ $\frac{d^2S}{dr^2} = 4\pi + \frac{360\pi}{r^3}$ $\frac{d^2S}{dr^2} = 4\pi + \frac{360\pi}{r^3} \Rightarrow \left(\frac{d^2S}{dr^2} = 37.699...\right)$ $37.699 > 0 \Rightarrow$ hence minimum	M1 M1 A1 M1 A1ft [5]
(c)	$S = 2\pi \times 3.556...^2 + \frac{2\times 90\pi}{3.556..} = ...$ $S = 238.4769... \Rightarrow S = 238\ (\text{cm}^2)$	M1 A1 [2]
		Total 10 marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 1 of 33 · Question 8 part b · 5 marks

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		$S = 2\pi r^2 + \frac{2\times 90\pi}{r} = 2\pi r^2 + \frac{180\pi}{r}$ Must have the $S =$ for this mark.
(b)	**M1**	For attempting to differentiate the given expression for S at least one power to decrease and neither power to increase. $\frac{dS}{dr} = 4\pi r - \frac{180\pi}{r^2}$
	M1	Sets their $\frac{dS}{dr} = 0$ and attempts to solve for r $4\pi r - \frac{180\pi}{r^2} = 0 \Rightarrow 4\pi r = \frac{180\pi}{r^2} \Rightarrow r^3 = 45 \Rightarrow r = ...$
	A1	For the correct value of $r = 3.55689... \Rightarrow r \approx 3.56$ Accept awrt 3.56
	M1	For attempting to differentiate their expression for $\frac{dS}{dr}$ at least one power to decrease and neither power to increase. $\frac{d^2S}{dr^2} = 4\pi + \frac{360\pi}{r^3}$
	A1ft	For correct work throughout $\frac{d^2S}{dr^2} = 4\pi + \frac{180\pi}{r^3} \Rightarrow \left(\frac{d^2S}{dr^2} = 37.699...\right)$ $37.699 > 0 \Rightarrow$ hence minimum Evaluation not required as both terms positive so $\frac{d^2S}{dr^2} > 0$ hence minimum Indication of positive or >0 required. If $\frac{d^2S}{dr^2}$ evaluated incorrectly then do not award. If evaluated then accept awrt 38
(c)	**M1**	For substituting their value of r into the **given** expression for S $S = 2\pi \times '3.556'...^2 + \frac{2\times 90\pi}{'3.556'..} = ...$ Their value of r >0
	A1	$S = 238.4769... \Rightarrow S = 238\ (\text{cm}^2)$ Accept awrt 238

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 1 of 33 · Question 8 part b · 5 marks

2022 JAN · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 2 of 33 · Question 9 part b · 5 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

Number		
9 (a)	$540 = 3x^2h \Rightarrow h = \frac{180}{x^2}$ $S = 2\left(3x^2 + 3xh + xh\right) = 6x^2 + 8xh$ $\Rightarrow S = 6x^2 + 8x \times \frac{180}{x^2} = 6x^2 + \frac{1440}{x}$ *	M1 M1 depM1A1 [4]
(b)	$S = 6x^2 + 1440x^{-1}$ $\frac{dS}{dx} = 12x - 1440x^{-2}$ At min/max $\frac{dS}{dx} = 0$ $12x - 1440x^{-2} = 0 \Rightarrow x^3 = 120 \Rightarrow x = 4.93242....$ $x \approx 4.93$ (3sf) $\frac{d^2S}{dx^2} = 12 + \frac{2880}{x^3} \Rightarrow$ Always positive for positive values of x, hence minimum	M1 M1A1 M1A1ft [5]
(c)	$S = 6 \times 4.93242^2 + \frac{1440}{4.93242} = 437.9185 \approx 438$	B1 [1]
		Total 10 marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 2 of 33 · Question 9 part b · 5 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 3 of 33 · Question 8 part b · 6 marks

2023 JAN · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

8 a	(Area) $= 2xy + \frac{1}{2}x^2\left(\frac{1}{2}\right)\left(= 2xy + \frac{1}{4}x^2 = 50\right)$	M1
	So $y = \frac{50 - \frac{1}{4}x^2}{2x}\left(= \frac{25}{x} - \frac{1}{8}x\right)$	dM1
	$P = 2x + 4y + \frac{1}{2}x = \frac{5}{2}x + 4y$	M1
	$= \frac{5}{2}x + 4\left(\frac{25}{x} - \frac{1}{8}x\right) \Rightarrow P = 2x + \frac{100}{x}$ *	M1 A1 cso (5)
b	$\left(\frac{dP}{dx} =\right) 2 - \frac{100}{x^2}$	M1 A1
	$\left(\frac{dP}{dx} = 0 \Rightarrow\right) 2 - \frac{100}{x^2} = 0 \Rightarrow x^2 = 50 \Rightarrow x = 5\sqrt{2}$ oe	M1 A1
	$\frac{d^2P}{dx^2} = \frac{200}{x^3}$ When $x = 5\sqrt{2}$ $\frac{d^2P}{dx^2} = \frac{2}{5}\sqrt{2}(= 0.565.....) > 0$ therefore minimum	M1 A1 (6)
c	$P = 2\left(5\sqrt{2}\right) + \frac{100}{5\sqrt{2}} = 20\sqrt{2}$	M1 A1 (2)
		Total 13 marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 3 of 33 · Question 8 part b · 6 marks

2023 JAN · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For writing an expression (may not be simplified) for the area in terms of x and y. Allow any expression of the form $xy + xy + ax^2$ oe where $a > 0$ This mark can be implied by an equation for A with the required $xy + xy + ax^2$ oe where $a > 0$
	dM1	For placing their expression for area = 50 and correctly rearranging to make y the subject. The expression for y does not need to be simplified.
	M1	For correctly writing an equation for perimeter in terms of x and y (may not be simplified). There must be a $P =$, though this may be seen later in their work. As long as $P =$ appears by the end of the work, this mark can be awarded. Perimeter = instead of $P =$ is acceptable for this mark
	M1	For correct substitution of their expression for y into their expression for **P or the Perimeter to reach an expression for P which must be in terms of x only. Simplification not required.**
	A1*cso	Obtains the given result with no errors. Perimeter = instead of $P =$ is **NOT** acceptable for this mark
For this question, if we are confident any second solution provided is a re-start and the work the candidate wishes us to consider, this is the work can be marked.		
(b)	M1	For an attempt to differentiate the given expression for P wrt x It is not necessary to see $\frac{dP}{dx} =$ Award this mark for any expression of the form $2 \pm \frac{b}{x^2}$, $b \neq 0$ oe
	A1	For $2 - \frac{100}{x^2}$ oe
	M1	For placing their expression for $\frac{dP}{dx} = 0$ which must have a term in x^2, a correct rearrangement to solve the equation formed and an attempt to find a value for x
	A1	For $x = 5\sqrt{2}$ oe ignore $x = -5\sqrt{2}$
	M1	For $\frac{d^2P}{dx^2} = \pm\frac{c}{x^3}$ $c \neq 0$
	A1	For substituting in a correct value for x, evaluating correctly, stating > 0 and drawing a conclusion. This can be as simple as # or shown. Students may also argue that the second derivative has to be positive as x is positive. This must be a convincing argument that since x is positive the derivative is positive and then a valid minimal conclusion. Do not award this mark if a negative value of x is used and it is not discounted at some point.
	ALT M1	Final 2 marks For substitution of valid values for x either side of their value of x found for the minimum value into $\frac{dP}{dx} = 2 \pm \frac{b}{x^2}$, $b \neq 0$
	A1	For fully correct substitution of their values for x into a correct expression for a first derivative, correctly evaluating or arguing the sign of the derivative and a conclusion drawn. Do not award this mark if a negative value of x is used and it is not discounted at some point.
(c)	M1	For substitution of their value for $x = 5\sqrt{2}$ oe into the given expression for P
	A1	For $20\sqrt{2}$

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 3 of 33 · Question 8 part b · 6 marks

2023 JAN · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 4 of 33 · Question 7 part b · 5 marks

2023 NOV · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

7(a)	$S = 5x^3 + (3x-4)^2$ $\Rightarrow S = 5x^3 + 9x^2 - 24x + 16$ *	M1 A1 cso [2]
(b)	$\frac{dS}{dx} = 15x^2 + 18x - 24 = 0$ $\Rightarrow (5x-4)(x+2) = 0 \Rightarrow x = \frac{4}{5}, -2$ $\frac{d^2S}{dx^2} = 30x + 18 = 30\left(\frac{4}{5}\right) + 18 \Rightarrow$ +ve hence minimum	M1 M1A1 M1A1 [5]
(c)	$S = 5\left(\frac{4}{5}\right)^3 + 9\left(\frac{4}{5}\right)^2 - 24\left(\frac{4}{5}\right) + 16 = \frac{128}{25}$ or 5.12	M1A1 [2]
		Total 9 marks

Part	Mark	Notes
(a)	M1	For substituting y into the given S Substituting x will not yield the required expression.
	A1 cso	For obtaining the given expression with no errors. You must check every line of their working.
(b)	M1	For an attempt to differentiate **the given** expression for S wrt x, Accept **at least** two terms fully correct with no power of x to increase.
	M1	Sets their differentiated expression = 0 and attempts to solve, **provided** it is a quadratic. See General Guidance for the definition of an attempt to solve a QE
	A1	For the correct **two** values of x.
	M1	Attempts to differentiate again. Minimally acceptable attempt is $\left(\frac{d^2S}{dx^2}\right) = Ax + B$
	A1	Conclusion: Concludes that the positive value of x $\left(\frac{4}{5}\right)$ will give a positive $\frac{d^2S}{dx^2}$ hence will be a minimum. For example, positive + positive = positive hence minimum. OR Substitutes either value of x, with the appropriate conclusion and correctly concludes that $x = \frac{4}{5}$ gives a minimum.

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 4 of 33 · Question 7 part b · 5 marks

2023 NOV · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

		NOTE: If they evaluate $\frac{d^2S}{dx^2}$ it must be correct [= 42] An incorrect value is A0.
(c)	M1	Uses **their** minimum value of x in the **given** expression, even if they identify $x = -2$ as the value giving a minimum S. If their minimum is not explicitly identified, then award M0. If they substitute both values without identifying which is which, award M0.
	A1	For the correct value of S

Question	Scheme	Marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 5 of 33 · Question 11 part b · 6 marks

2023 JAN · Paper 02 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

11	$S = \pi r^2 + 2\pi rh$ $\pi r^2 + 2\pi rh = 625\pi \Rightarrow 2rh = 625 - r^2 \Rightarrow h = \dfrac{625 - r^2}{2r}$ *	B1 M1A1 [cso] [3]
(b)	$V = \pi r^2\left(\dfrac{625 - r^2}{2r}\right)$ or $\left(V = \dfrac{625\pi}{2}r - \dfrac{\pi r^3}{2}\right)$ $\dfrac{dV}{dr} = \dfrac{625\pi}{2} - \dfrac{3\pi r^2}{2}$ $\dfrac{625\pi}{2} - \dfrac{3\pi r^2}{2} = 0 \Rightarrow 625 = 3r^2 \Rightarrow r = 14.4337... \Rightarrow r \approx 14.4$ For differentiating their $\dfrac{dV}{dr}$ $\dfrac{d^2V}{dr^2} = -3\pi r \ (= -136.03\ ...)$ For concluding that as $\dfrac{d^2V}{dr^2}$ is negative then the volume is a maximum. (Explicit substitution of r is not required).	M1 M1 M1A1 M1 A1 [6]
(c)	$h = \dfrac{625 - 14.43375^2}{2 \times 14.43375} = 14.4337... \approx 14.4$	B1ft [1]
		Total 10 marks

Question	Scheme	Marks
11	For a correct expression for the surface area of the cup. $S = \pi r^2 + 2\pi rh$	B1
	For equating their expression for the surface area to 625π and attempting to make h the subject. $\pi r^2 + 2\pi rh = 625\pi \Rightarrow 2rh = 625 - r^2 \Rightarrow h = \dfrac{625 - r^2}{2r}$ Allow errors in arithmetic but not mathematically incorrect process.	M1
	For the correct expression for h as shown with no errors. $h = \dfrac{625 - r^2}{2r}$*	A1 cso [3]

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 5 of 33 · Question 11 part b · 6 marks

2023 JAN · Paper 02 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

(b)	For the correct volume of the container in terms of r only. $V = \pi r^2\left(\frac{625-r^2}{2r}\right)$ or $\left(V = \frac{625\pi}{2}r - \frac{\pi r^3}{2}\right)$	M1
	Minimally acceptable attempt at differentiation, see general guidance, no power to increase. $\frac{dV}{dr} = \frac{625\pi}{2} - \frac{3\pi r^2}{2}$	M1
	Places their derivative = 0 and attempts to rearrange to find r. Minimally acceptable derivative is of the form $a \pm b\pi r^2$ $\frac{625\pi}{2} - \frac{3\pi r^2}{2} = 0 \Rightarrow 625 = 3r^2 \Rightarrow r = ...$	M1
	For the correct value of r $r = 14.4337... \Rightarrow r \approx 14.4$ (cm) Must reject negative value if found, award A0 if not rejected. May be implied from subsequent working.	A1
	For differentiating their $\frac{dV}{dr}$ For minimally acceptable attempt to differentiate their first derivative, see general guidance, no power to increase. Or testing gradients or a sketch. $\frac{d^2V}{dr^2} = -3\pi r$	M1
	For concluding that as $\frac{d^2V}{dr^2}$ is negative then the volume is a maximum. (Explicit substitution of r is not required).	A1 [6]
(c)	For the value of h using their value of r $h = \frac{625-14.43375^2}{2\times 14.43375} = 14.4337... \approx 14.4$ Allow for 14.5 if substituting $r = 14.4$	B1ft [1]
		Total 10 marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 5 of 33 · Question 11 part b · 6 marks

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Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 6 of 33 · Question 7 part b · 7 marks

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7 (a)	$2x^2y$	M1
	$72 = 2x^2y \Rightarrow y = \frac{36}{x^2}$ oe or $xy = \frac{36}{x}$ oe	M1
	$(A =) 2\times 2xy + 2\times xy + 2\times 2x^2 \left(= 6xy + 4x^2\right)$	
	$= 6x\left(\frac{36}{x^2}\right) + 4x^2$ or $= 6\left(\frac{36}{x}\right) + 4x^2$	dM1
	$A = \frac{216}{x} + 4x^2$ *	A1*cso [4]
(b)	$\left(\frac{dA}{dx} =\right) -\frac{216}{x^2} + 8x$	M1
	$0 = -\frac{216}{x^2} + 8x \rightarrow x =$ or $x^3 =$	dM1
	$x = 3$	A1
	$(A =) \frac{216}{3} + 4\times 3^2$	dM1
	$= 108$	A1
	$\left(\frac{d^2A}{dx^2} =\right) \frac{432}{x^3} + 8$	M1
	$= \frac{432}{3^3} + 8 (= 24) > 0$ therefore minimum	A1 [7]
		Total 11 marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 6 of 33 · Question 7 part b · 7 marks

2025 JUN · Paper 02 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	Correct expression for volume.
	M1	Equates their expression of the form ax^2y to 72 and rearranges correctly for y or xy Simplification is not required to be awarded this mark.
	dM1	Substitutes their expression for y or xy into a correct (unsimplified) formula or expression for area.. Dep on previous M mark
	A1*	Correct solution only, minimum steps as shown. A = must be present at least once.
(b)	M1	Differentiates to get an expression of the form $\pm\frac{a}{x^2}+8x \quad a\neq 0$
	dM1	Places their expression of the correct form = 0 and rearranges to give x or x^3 = Allow errors in rearrangement. Dependent on previous method mark. So long as differentiation shown, this mark can be implied by correct value.
	A1	$x = 3$ dependent on having gained 1st method mark – the question directs the candidates to use calculus.
	dM1	Substitutes their value for x into the **correct** expression for A. Dependent on 1st method mark. So long as differentiation shown, this mark can be implied by a correct value.
	A1	For 108 dependent on having gained 1st method mark – the question directs the candidates to use calculus.
	M1	Differentiates their expression for $\frac{dA}{dx}$ to give an expression of the form $\pm\frac{b}{x^3}+8 \; b\neq 0$
	A1	Correctly substitutes $x = 3$ into the correct second derivative or argues the correct second derivative is > 0 and draws a conclusion. It is not necessary to see an evaluation, but the evaluation must be correct if carried out. Second derivative **must be stated to be > 0 or positive.** As always with A1, must have gained all previous marks. **Note: it is highly unusual to have a final A1, when any of the preceding marks have been 0. The structure of this question means, this is possible, if they either don't find the minimum area or calculate it incorrectly.** **So for the final 4 marks dM1 A0 M1 A1 and dM0 A0 M1 A1**

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Question number	Scheme	Marks

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2025 JUN · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

6 (a)	$700\pi = 2\pi r^2 + 2\pi rh$	B1
	$h = \dfrac{350 - r^2}{r}$ or $rh = 350 - r^2$	M1
	$V = \pi r^2 h = \pi r^2\left(\dfrac{350 - r^2}{r}\right) = \left\{\pi r\left(350 - r^2\right)\right\} = \pi r\left(350 - r^2\right)*$	M1A1* cso [4]
(b)	$V = 350\pi r - \pi r^3$	
	$\dfrac{dV}{dr} = 350\pi - 3\pi r^2 = 0$	M1
	$\Rightarrow 350\pi = 3\pi r^2 \Rightarrow r = \sqrt{\dfrac{350}{3}} = 10.8$ (3sf)	M1A1
	$\dfrac{d^2V}{dr^2} = -6\pi r$	M1
	$\dfrac{d^2V}{dr^2} < 0 \therefore$ max	A1 [5]
(c)	$h = \dfrac{350 - r^2}{r} = \dfrac{350 - \dfrac{350}{3}}{\sqrt{\dfrac{350}{3}}} = 21.6$ (3 s.f.)	B1 [1]
		Total 10 marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 7 of 33 · Question 6 part b · 5 marks

2025 JUN · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Additional Guidance
(a)	B1	For a **correct** equation for the surface area in terms of the radius and height
	M1	For rearranging to find h in terms of r which must be as a minimum $h = \frac{k - lr^2}{mr}$ where k, l and m are integers OR $rh = k - lr^2$ where k and l are integers
	M1	For a correct expression for volume using *their* h in terms of r
	A1* cso	For rearranging and obtaining the given result with no errors or omissions and must include $V = \ldots$ [Allow Volume = …. or Vol = …..]
(b)	M1	For an attempt to differentiate the **given** expression for V which must be of the form $\frac{dV}{dr} = 350\pi - K\pi r^2$ where K is a constant Using Product Rule: $\frac{dV}{dr} = \pi r(-2r) + \pi(350 - r^2) \Rightarrow \frac{dV}{dr} = \left[350\pi - K\pi r^2\right]$
	M1	For setting their $\frac{dV}{dr} = 0$ **and** attempting to find a value for r.
	A1	$r = \sqrt{\frac{350}{3}}$ [=10.801...] accept awrt 10.8
	M1	For finding $\frac{d^2V}{dr^2}$ Accept only $\frac{d^2V}{dr^2} = \pm k\pi r$ where k is a constant See general guidance for what constitutes an attempt to differentiate.
	A1	For the correct conclusion derived from the correct working throughout.
(c)	B1	For obtaining the value of h correct to 3 s.f. Accept awrt 21.6

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11 (a)	Let x = the length of the side of the triangle and h = the length of the prism	
	$\frac{1}{2}x^2 \sin 60\, h = 72$ or $\frac{1}{2}\left(\sqrt{\left(x^2 - \left(\frac{1}{2}x\right)^2\right)}\right)xh = 72$	M1
	$\frac{\sqrt{3}x^2h}{4} = 72$	M1
	$h = \frac{288}{\sqrt{3}x^2}$	A1
	$S = 2 \times \frac{1}{2}x^2 \sin 60 + 3xh$ or $2 \times \frac{1}{2}\left(\sqrt{\left(x^2 - \left(\frac{1}{2}x\right)^2\right)}\right)x + 3xh$	M1
	$S = \frac{\sqrt{3}x^2}{2} + 3x\left(\frac{288}{\sqrt{3}x^2}\right)$	M1
	$S = \frac{\sqrt{3}x^2}{2} + \frac{288\sqrt{3}}{x}$ *	A1 cso (6)
(b)	$\frac{dS}{dx} = \sqrt{3}x - \frac{288\sqrt{3}}{x^2} (= 0)$	M1
	$x^3 = 288$	dM1
	$x = \sqrt[3]{288} = 6.604$	A1
	$\frac{d^2S}{dx^2} = \sqrt{3} + \frac{576\sqrt{3}}{x^3}$	ddM1
	$\frac{d^2S}{dx^2} > 0$ (when x =6.6) ∴ value is a minimum	A1 (5)
(c)	Substitutes their x into $S = \frac{\sqrt{3}x^2}{2} + \frac{288\sqrt{3}}{x}$	M1
	$S = 113$	A1 (2)
		[13]

Part	Mark	Additional Guidance
(a)	M1	For the correct expression for the volume of the prism in terms of x and h (or other letter for the length, e.g. l) Simplification not required for this mark $72 = \left(\frac{1}{2} \times x \times x \times \sin 60^\circ\right) \times h$ or $72 = \left(\frac{1}{2} \times x \times \sqrt{x^2 - \frac{x^2}{4}}\right) \times h$ or $72 = \left(\frac{\sqrt{3}}{4}x^2\right) \times h$
	M1	For an attempt to find an expression for h in terms of x Accept as a minimum $h = \frac{k}{x^2}$ where k is a positive integer
	A1	For $h = \frac{288}{\left(\sqrt{3}\right)x^2}$ or $h = \frac{96\sqrt{3}}{x^2}$

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Answer 8 of 33 · Question 11 part b · 5 marks

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	M1	For an expression for S in terms of x and h (ft their area of the triangle) $S = 2\left(\frac{1}{2}\times x\times x\times \sin 60^\circ\right)+3xh$ or $S = 2\left(\frac{1}{2}\times x\times \sqrt{x^2-\frac{x^2}{4}}\right)+3xh$ $\left(S=\frac{\sqrt{3}}{2}x+3xh\right)$
	M1	For substituting their h into their S $S = 2\left(\frac{1}{2}\times x\times x\times \sin 60^\circ\right)+\left(3x\times\frac{288}{\left(\sqrt{3}\right)x^2}\right)$ or $S = 2\left(\frac{1}{2}\times x\times \sqrt{x^2-\frac{x^2}{4}}\right)+\left(3x\times\frac{288}{\left(\sqrt{3}\right)x^2}\right)$
	A1	For the correct expression for S as given. **The expression must be set equal to S.** $S=\frac{\sqrt{3}x^2}{2}+\frac{288\sqrt{3}}{x}$ exactly as seen here. * This is a given result so full working must be seen.
(b)	M1	For an attempt to differentiate the **given** expression for S $\frac{dS}{dx}=\sqrt{3}x-\frac{288\sqrt{3}}{x^2}$ or $\frac{dS}{dx}=\sqrt{3}x-288\sqrt{3}x^{-2}$ (See General Guidance for the definition of an attempt)
	dM1	For setting their differentiated expression = 0 and attempting to solve for x $\sqrt{3}x-\frac{288\sqrt{3}}{x^2}=0\Rightarrow x^3=288\Rightarrow x=\left(6.604\right)$ (rounded correctly) This mark is dependent on the first M mark in (b)
	A1	For $x = 6.604$ rounded correctly
	dM1	For attempting the second derivative (usual definition of an attempt) $\frac{d^2S}{dx^2}=\sqrt{3}+\frac{576\sqrt{3}}{x^3}$ This mark is dependent on first M mark in (b)
	A1ft	Concludes either that $\frac{d^2S}{dx^2}>0$ for all **positive** values of x **or** substitutes in their value of x to show that $\frac{d^2S}{dx^2}=5.19.....$ hence positive so must be a minimum. Only ft if the final conclusion is a minimum provided their $\frac{d^2S}{dx^2}$ is algebraically correct
	ALT – for justifying the minimum using their derivative	
	dM1	Chooses a value either side of their value of x and substituting them into their $\frac{dS}{dx}$ e.g. $x = 6$ and 7 $\frac{dS}{dx}=\sqrt{3}\times 6-\frac{288\sqrt{3}}{6^2}=-3.46...$ and $\frac{dS}{dx}=\sqrt{3}\times 7-\frac{288\sqrt{3}}{7^2}=1.944...$
	A1ft	Concludes that the gradient function moves from negative to positive hence must be a minimum.
(c)	M1	Substitutes their value of x into the given expression for S $S=\frac{\sqrt{3}\,'6.604'^2}{2}+\frac{288\sqrt{3}}{'6.604'}=...$
	A1	For $S = 113$ rounded correctly

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 8 of 33 · Question 11 part b · 5 marks

2020 JAN · Paper 01 · extract 3 of 5 · find part b in this mark scheme extract

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 8 of 33 · Question 11 part b · 5 marks

2020 JAN · Paper 01 · extract 4 of 5 · find part b in this mark scheme extract

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 8 of 33 · Question 11 part b · 5 marks

2020 JAN · Paper 01 · extract 5 of 5 · find part b in this mark scheme extract

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Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 9 of 33 · Question 5 part b.i · 0 marks

2023 JUN · Paper 01 · extract 1 of 3 · find part b.i in this mark scheme extract

5(a)	$V = x \times 4x \times h \Rightarrow 4x^2h = 75 \Rightarrow h = \frac{75}{4x^2}$ $S = 2\left(4x^2 + xh + 4xh\right) = \left[8x^2 + 2xh + 8xh\right] \Rightarrow S = 8x^2 + 10xh$ $\Rightarrow S = 8x^2 + 10x \times \frac{75}{4x^2} \Rightarrow S = 8x^2 + \frac{375}{2x}$ *	B1 M1 M1A1 cso [4]
(b)(i)	$\frac{dS}{dx} = 16x - \frac{375}{2x^2} = 0$ $\Rightarrow 16x = \frac{375}{2x^2} \Rightarrow x^3 = \frac{375}{32} \Rightarrow x = 2.2714... \approx 2.27\ (\text{cm})$	M1 M1A1
(ii)	$\frac{d^2S}{dx^2} = 16 + 375x^{-3} \approx 48$ Conclusion: (As x is always positive) $\frac{d^2S}{dx^2} > 0 \Rightarrow$ minimum	M1 A1ft [5]
(c)	$S = 8 \times '2.27'^2 + \frac{375}{2 \times '2.27'} = 123.822... \approx 124\ (\text{cm}^2)$	M1A1 [2]
		Total 11 marks

Part	Mark	Notes
(a)	**B1**	For finding the correct expression for h in terms of x Accept this mark seen anywhere in part (a) Accept also for example, $xh = \frac{75}{4x}$ or any equivalent that can be substituted into an expression for S.
	M1	For finding an expression for S is terms of x and h. Accept as a minimum $Ax^2 + Bxh$ where A and B are constants. Accept other letters for S at this stage. E.g SA etc. or even no letter at all.
	M1	For substituting their h into their S
	A1	For the correct expression for S as shown in the question. You must see $S = 8x^2 + \frac{375}{2x}$ including S although S seen at the top of a column of working is also fine.
(b)	**M1**	For an attempt to differentiate the given S which as a minimum must be of the form

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 9 of 33 · Question 5 part b.i · 0 marks

2023 JUN · Paper 01 · extract 2 of 3 · find part b.i in this mark scheme extract

		$\frac{dS}{dx} = 16x - \frac{k}{x^2}$ where k is a constant.
	M1	For setting their $\frac{dS}{dx} = 0$ and attempting to solve the equation and reaching a value of x. It must be a clearly differentiated expression even if they do not score the first M mark in (b). That is, they must reduce the power of at least one term by 1.
	A1	For awrt $x = 2.27$
	M1	For an attempt to differentiate their $\frac{dS}{dx}$ to find $\frac{d^2S}{dx^2}$ which must be as a minimum $\frac{d^2S}{dx^2} = 16 + lx^{-3}$ where l is a constant **Other methods** Please send to Review any examples of candidates who test the gradient on either side of the minimum, or who draw a sketch.
	A1ft	Concludes that as $x > 0$ so $\frac{d^2S}{dx^2} > 0$ and therefore S is a minimum when $x = 2.27$ Dependent on correct method seen. Follow through their value of x provided it is positive. **NB:** They do not need to evaluate $\frac{d^2S}{dx^2}$ to score this mark. [See above] However, if they use the correct [2.27] or an incorrect positive value of x, provided substitution is seen award the mark even if the final evaluation is incorrect. If no substitution is seen and either the value of x is incorrect or they do not obtain approximately 48, withhold this mark.
(c)	**M1**	Substitutes their $x = 2.27$ into the given S Follow through their value of x provided it is a positive value. Use of a negative value of x is M0. If the final value of S is incorrect following an incorrect value of x award this mark only for explicit substitution seen.
	A1	For awrt $S = 124$ (cm^2)

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 9 of 33 · Question 5 part b.i · 0 marks

2023 JUN · Paper 01 · extract 3 of 3 · find part b.i in this mark scheme extract

Question	Scheme	Marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 10 of 33 · Question 5 part b.ii · 5 marks

2023 JUN · Paper 01 · extract 1 of 3 · find part b.ii in this mark scheme extract

5(a)	$V = x \times 4x \times h \Rightarrow 4x^2h = 75 \Rightarrow h = \frac{75}{4x^2}$ $S = 2\left(4x^2 + xh + 4xh\right) = \left[8x^2 + 2xh + 8xh\right] \Rightarrow S = 8x^2 + 10xh$ $\Rightarrow S = 8x^2 + 10x \times \frac{75}{4x^2} \Rightarrow S = 8x^2 + \frac{375}{2x}$ *	B1 M1 M1A1 cso [4]
(b)(i)	$\frac{dS}{dx} = 16x - \frac{375}{2x^2} = 0$ $\Rightarrow 16x = \frac{375}{2x^2} \Rightarrow x^3 = \frac{375}{32} \Rightarrow x = 2.2714... \approx 2.27 \ (\text{cm})$	M1 M1A1
(ii)	$\frac{d^2S}{dx^2} = 16 + 375x^{-3} \approx 48$ Conclusion: (As x is always positive) $\frac{d^2S}{dx^2} > 0 \Rightarrow$ minimum	M1 A1ft [5]
(c)	$S = 8 \times '2.27'^2 + \frac{375}{2 \times '2.27'} = 123.822... \approx 124 \ (\text{cm}^2)$	M1A1 [2]
		Total 11 marks

Part	Mark	Notes
(a)	**B1**	For finding the correct expression for h in terms of x Accept this mark seen anywhere in part (a) Accept also for example, $xh = \frac{75}{4x}$ or any equivalent that can be substituted into an expression for S.
	M1	For finding an expression for S is terms of x and h. Accept as a minimum $Ax^2 + Bxh$ where A and B are constants. Accept other letters for S at this stage. E.g SA etc. or even no letter at all.
	M1	For substituting their h into their S
	A1	For the correct expression for S as shown in the question. You must see $S = 8x^2 + \frac{375}{2x}$ including S although S seen at the top of a column of working is also fine.
(b)	**M1**	For an attempt to differentiate the given S which as a minimum must be of the form

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 10 of 33 · Question 5 part b.ii · 5 marks

2023 JUN · Paper 01 · extract 2 of 3 · find part b.ii in this mark scheme extract

		$\frac{dS}{dx} = 16x - \frac{k}{x^2}$ where k is a constant.
	M1	For setting their $\frac{dS}{dx} = 0$ and attempting to solve the equation and reaching a value of x. It must be a clearly differentiated expression even if they do not score the first M mark in (b). That is, they must reduce the power of at least one term by 1.
	A1	For awrt $x = 2.27$
	M1	For an attempt to differentiate their $\frac{dS}{dx}$ to find $\frac{d^2S}{dx^2}$ which must be as a minimum $\frac{d^2S}{dx^2} = 16 + lx^{-3}$ where l is a constant **Other methods** Please send to Review any examples of candidates who test the gradient on either side of the minimum, or who draw a sketch.
	A1ft	Concludes that as $x > 0$ so $\frac{d^2S}{dx^2} > 0$ and therefore S is a minimum when $x = 2.27$ Dependent on correct method seen. Follow through their value of x provided it is positive. **NB:** They do not need to evaluate $\frac{d^2S}{dx^2}$ to score this mark. [See above] However, if they use the correct [2.27] or an incorrect positive value of x, provided substitution is seen award the mark even if the final evaluation is incorrect. If no substitution is seen and either the value of x is incorrect or they do not obtain approximately 48, withhold this mark.
(c)	**M1**	Substitutes their $x = 2.27$ into the given S Follow through their value of x provided it is a positive value. Use of a negative value of x is M0. If the final value of S is incorrect following an incorrect value of x award this mark only for explicit substitution seen.
	A1	For awrt $S = 124$ (cm^2)

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 10 of 33 · Question 5 part b.ii · 5 marks

2023 JUN · Paper 01 · extract 3 of 3 · find part b.ii in this mark scheme extract

Question	Scheme	Marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 11 of 33 · Question 8 part b · 4 marks

2024 JUN · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

8(a)	Finds the length DE $x^2 = (DE)^2 + (DE)^2 \Rightarrow DE = \frac{x}{\sqrt{2}}$ or $DE^2 = \frac{x^2}{2}$ Volume of prism $3.6 = \left(\frac{x}{\sqrt{2}}\right)^2 \times \frac{1}{2} \times y \Rightarrow \left[y = \frac{72}{5x^2}\right]$ Surface area of prism $S = 2 \times \frac{x^2}{4} + 2 \times \frac{x}{\sqrt{2}} \times \frac{72}{5x^2} + x \times \frac{72}{5x^2} = \left[\frac{x^2}{2} + \sqrt{2} \times \frac{72}{5x} + \frac{72}{5x}\right]$ $\Rightarrow S = \frac{x^2}{2} + \frac{72(\sqrt{2}+1)}{5x}$ *	B1 M1 M1 A1 cso [4]
(b)	$\frac{dS}{dx} = x - \frac{72(\sqrt{2}+1)}{5x^2} = 0$ $\Rightarrow \frac{72(\sqrt{2}+1)}{5x^2} = x$ $\Rightarrow x = \sqrt[3]{\frac{72(\sqrt{2}+1)}{5}} = 3.26371\ldots \approx 3.26\ (\text{cm})$ $\frac{d^2S}{dr^2} = 1 + \frac{144(\sqrt{2}+1)}{5x^3}$ $\Rightarrow$ positive + positive = positive $\Rightarrow$ is always greater than 0, hence minimum	M1 M1 A1 B1ft [4]
(c)	$S_{(\min)} = \frac{3.26371^2}{2} + \frac{72(\sqrt{2}+1)}{5 \times 3.26371} = 15.97779\ldots \approx 16\ (\text{cm}^2)$	M1A1 [2]
		Total 10 marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 11 of 33 · Question 8 part b · 4 marks

2024 JUN · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

(a)	B1	For using Pythagoras theorem or any appropriate trigonometry to find the length DE (or any equivalent side)
	M1	For using the correct formula to find the volume of a prism and rearranging to find an expression for y minimally of the form $y = \frac{A}{Bx^2}$ This need not be simplified.
	M1	For forming an expression for S using their y The basic expression for the surface area must be correct, with correct substitution of their values. $S = 2\times'DE'+2\times'DE'\times y+xy \Rightarrow S = 2\times'\frac{x^2}{4}'+2\times'\frac{x}{\sqrt{2}}'\times\left('\frac{72}{5x^2}'\right)+x\times\left('\frac{72}{5x^2}'\right)$
	A1 cso	For $S = \frac{x^2}{2}+\frac{72\left(\sqrt{2}+1\right)}{5x}$ as written
(b)	M1	For differentiating the **given** expression for S Minimally acceptable differentiated expression is as follows: $\frac{dS}{dx} = x \pm Kx^{-2}$ oe. where K is a constant
	M1	For setting their $\frac{dS}{dx} = 0$ and attempting to solve to obtain a value for x
	A1	For the correct value of x Accept awrt 3.26 (cm)
	B1ft	For differentiating their $\frac{dS}{dr}$ to minimally obtain $\frac{d^2S}{dr^2} = 1 \pm Lx^{-3}$ and coming to an appropriate conclusion. The justification has to be correct and complete. If they refer/use a substitution it must be seen and it must be correct.. That is, $\frac{d^2S}{dr^2} = 1+\frac{144\left(\sqrt{2}+1\right)}{5\times3.26^3} = 3.00...$ $3 > 0$ so positive, hence minimum Alternatively: x is positive so; positive $+ \frac{\text{positive}}{\text{positive}} \Rightarrow$ positive hence minimum
(c)	M1	For substituting their value of x (provided it is positive) and attempting to find a value of S
	A1	For awrt $S = 16$

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 11 of 33 · Question 8 part b · 4 marks

2024 JUN · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 12 of 33 · Question 5 part b · 5 marks

2025 NOV · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

5(a)	$\left[S = 2\pi rh + 2\pi r^2 \Rightarrow 322\pi = 2\pi rh + 2\pi r^2\right]$ $\Rightarrow (h =)\dfrac{161 - r^2}{r}$ oe eg $\dfrac{161}{r} - r$ eg $\dfrac{322\pi - 2\pi r^2}{2\pi r}$ $\left(V = \pi r^2 h \Rightarrow\right)(V =)\pi r^2\left("\dfrac{161 - r^2}{r}"\right) \Rightarrow$ $V = \pi r\left(161 - r^2\right)$ *	B1 M1 A1cso [3]
ALT	$\left(S = 2\pi rh + 2\pi r^2 \Rightarrow 322\pi = 2\pi rh + 2\pi r^2 \Rightarrow 161 = rh + r^2\right)$ $\Rightarrow (rh =) 161 - r^2$ $\left(V = \pi r^2 h \Rightarrow\right)(V =)\pi r\left(161 - r^2\right)$ *	B1 M1 A1
(b)	$\left[\dfrac{dV}{dr} =\right] 161\pi - 3\pi r^2$ oe $161\pi - 3\pi r^2 = 0 \Rightarrow r^2 = \dfrac{161}{3} \Rightarrow r = 7.3257... \Rightarrow r \approx 7.33$ $\left[\dfrac{d^2V}{dr^2} =\right] - 6\pi r = [[-138.17, -138.09]]$ or -138 or -140 or awrt -44π < 0 hence maximum	M1 M1A1 M1A1 [5]
(c)	$V = \pi \times 7.3257\left(161 - 7.3257^2\right) = 2470$	M1A1 [2]
		Total 10 marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 12 of 33 · Question 5 part b · 5 marks

2025 NOV · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	B1	For finding a correct expression for h in terms of r This expression doesn't need to be simplified. Independent of method mark, so mark the expression for h in terms of r for this mark. For the final A mark, must be no errors.
	M1	For substituting their expression for h in terms of r into a correct expression for V $V =$ doesn't need to be present for this mark.
	A1* cso	For the correct formula exactly as written. $V =$ must be shown at least once, throughout working. Minimum steps as shown in the mark scheme, no errors or omissions. Steps shown in square brackets are not required, but are provided as if present, must be correct and must be checked by Examiners. All additional steps must also be checked.
ALT	B1	For finding a correct expression for rh Independent of method mark, so mark the expression for rh in terms of r for this mark. For the final A mark, must be no errors.
	M1	For substituting their expression for rh in terms of r into correct expression for V $V =$ doesn't need to be present for this mark.
	A1* cso	For the correct formula exactly as written. $V =$ must be shown at least once, throughout working. Minimum steps as shown in the mark scheme, no errors or omissions. Steps shown in square brackets are not required, but are provided as if present, must be correct and must be checked by Examiners. All additional steps must also be checked.
(b)	M1	For a minimally acceptable attempt (see general guidance, but also no power or r to increase) to differentiate the given expression. Simplification is not necessary.
	M1	Sets their derivative = 0 and **correctly** rearranges to finds a value for r or r^2 Allow 7.3 without a full rearrangement to imply this mark. This isn't a completely dependent mark, but must follow from anything considered to be an attempt to differentiate and then rearrange an equation with a 2 term expression in r Ignore any value of negative r
	A1	For the correct value of r. Accept awrt 7.33 Ignore any value of negative r Because candidates are told to use calculus, a correct answer will not imply method marks.
	M1	For a minimally acceptable attempt (see general guidance, but also no power or r to increase) to differentiate their expression for the first derivative which must have 2 terms, with a least one term in r
	A1	For a correct conclusion. The value of the second derivative doesn't have to be calculated, but if it is, the numbers in the interval, or 138 or 140 must be seen. Must state < 0 and a brief conclusion which could be as short as shown or # Don't award if there is work to consider a negative value of r, unless discounted or abandoned. **All work must be correct throughout (b) for this mark.** Must be a clear conclusion referring to only the positive value of r for this mark.
(c)	M1	For substituting their value of r into the given expression for V May be implied by a correct value.
	A1	awrt 2470 but not from 7.3 or incorrect r

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 12 of 33 · Question 5 part b · 5 marks

2025 NOV · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 13 of 33 · Question 11 part b.i · 0 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 3 · find part b.i in this mark scheme extract

Number		
11 (a)	$(600 =)2\pi r^2 + 2\pi rh$ oe eg $(300 =)\pi r^2 + \pi rh$	M1
	$h = \frac{300-\pi r^2}{\pi r}$ oe | $\pi rh = 300 - \pi r^2$ oe	A1 cao
	$(V =)\pi r^2 "\left(\frac{300-\pi r^2}{\pi r}\right)"$ oe | $V = "(300 - \pi r^2)"r$ oe	M1
	$V = 300r - \pi r^3$ *	A1* cso [4]
(b) (i)	$\frac{dV}{dr} = 300 - 3\pi r^2$	M1
	$0 = 300 - 3\pi r^2 \quad \rightarrow \quad r =$	M1
	$r = \sqrt{\frac{100}{\pi}}$ * cso	A1* cso
(ii)	$\frac{d^2V}{dr^2} = -6\pi r$	M1
	$\rightarrow \frac{d^2V}{dr^2} = -6\pi\sqrt{\frac{100}{\pi}}$ or $\frac{d^2V}{dr^2} = -6\pi \times 5.6418958.....$	
	When r is positive, $-6\pi r$ is negative (-106.347231…..) and therefore this value of r gives a maximum	A1 [5]
(c)	$(V =)300"\sqrt{\frac{100}{\pi}}" - \pi\left("\sqrt{\frac{100}{\pi}}"\right)^3 (= 1128.(379167))$	M1
	$p^3 = \frac{300"\sqrt{\frac{100}{\pi}}" - \pi\left("\sqrt{\frac{100}{\pi}}"\right)^3}{\frac{4}{3}\pi}$ (=269(.3806…))	dM1
	$p = 6.5$ cm	A1 [3]
		Total 12 marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 13 of 33 · Question 11 part b.i · 0 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 3 · find part b.i in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	Correct expression for the surface area of the cylinder and an attempt to rearrange to $h =$ or $\pi rh =$ Allow errors in arithmetic but not mathematically incorrect process. πrh may be embedded, e.g. $300 = \pi r^2 + \pi rh$ becoming $300 = \pi r^2 + V$ would score M1A1M1 and may score full marks if correct final result obtained.
	A1	cao
	M1	Substitutes their expression for height or their expression for πrh into a correct expression for the volume.
	A1	cso no errors or omissions, must state $V =$
	Mark parts (i) and (ii) together.	
(b) (i)	M1	Minimally acceptable attempt at differentiation, see general guidance, no power to increase.
	M1	Places their derivative = 0 and attempts to rearrange to find r. Minimally acceptable derivative is of the form $a \pm b\pi r^2$
	A1	Correct value for r, exact value only. Must reject negative value if found, award A0 if not rejected.
(ii)	M1	Minimally acceptable attempt to differentiate their first derivative, see general guidance, no power to increase. Or testing gradients or a sketch.
	A1	Correct evaluation of second derivative or explanation of why second derivative is negative. Conclusion this value of r gives a maximum. No incorrect work.
(c)	M1	Correct substitution of their r into the expression for V.
	dM1	Attempts rearrangement using the formula for volume of a sphere to make p^3 the subject Correct order of operations applied to right hand side. Accept arithmetic slips.
	A1	$p = 6.5$ Accept awrt 6.5

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 13 of 33 · Question 11 part b.i · 0 marks

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Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 14 of 33 · Question 11 part b.ii · 5 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 3 · find part b.ii in this mark scheme extract

number		
11 (a)	$(600 =)2\pi r^2 + 2\pi rh$ oe eg $(300 =)\pi r^2 + \pi rh$	M1
	$h = \frac{300 - \pi r^2}{\pi r}$ oe | $\pi rh = 300 - \pi r^2$ oe	A1 cao
	$(V =)\pi r^2 "\left(\frac{300 - \pi r^2}{\pi r}\right)"$ oe | $V = "(300 - \pi r^2)"r$ oe	M1
	$V = 300r - \pi r^3$ *	A1* cso [4]
(b) (i)	$\frac{dV}{dr} = 300 - 3\pi r^2$	M1
	$0 = 300 - 3\pi r^2 \quad \rightarrow \quad r =$	M1
	$r = \sqrt{\frac{100}{\pi}}$ * cso	A1* cso
(ii)	$\frac{d^2V}{dr^2} = -6\pi r$	M1
	$\rightarrow \frac{d^2V}{dr^2} = -6\pi\sqrt{\frac{100}{\pi}}$ or $\frac{d^2V}{dr^2} = -6\pi \times 5.6418958.....$	
	When r is positive, $-6\pi r$ is negative (-106.347231…..) and therefore this value of r gives a maximum	A1 [5]
(c)	$(V =)300"\sqrt{\frac{100}{\pi}}" - \pi\left("\sqrt{\frac{100}{\pi}}"\right)^3$ (= 1128. (379167))	M1
	$p^3 = \frac{300"\sqrt{\frac{100}{\pi}}" - \pi\left("\sqrt{\frac{100}{\pi}}"\right)^3}{\frac{4}{3}\pi}$ (=269(.3806…))	dM1
	p = 6.5 cm	A1 [3]
		Total 12 marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 14 of 33 · Question 11 part b.ii · 5 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 3 · find part b.ii in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	Correct expression for the surface area of the cylinder and an attempt to rearrange to $h =$ or $\pi rh =$ Allow errors in arithmetic but not mathematically incorrect process. πrh may be embedded, e.g. $300 = \pi r^2 + \pi rh$ becoming $300 = \pi r^2 + V$ would score M1A1M1 and may score full marks if correct final result obtained.
	A1	cao
	M1	Substitutes their expression for height or their expression for πrh into a correct expression for the volume.
	A1	cso no errors or omissions, must state $V =$
	Mark parts (i) and (ii) together.	
(b) (i)	M1	Minimally acceptable attempt at differentiation, see general guidance, no power to increase.
	M1	Places their derivative = 0 and attempts to rearrange to find r. Minimally acceptable derivative is of the form $a \pm b\pi r^2$
	A1	Correct value for r, exact value only. Must reject negative value if found, award A0 if not rejected.
(ii)	M1	Minimally acceptable attempt to differentiate their first derivative, see general guidance, no power to increase. Or testing gradients or a sketch.
	A1	Correct evaluation of second derivative or explanation of why second derivative is negative. Conclusion this value of r gives a maximum. No incorrect work.
(c)	M1	Correct substitution of their r into the expression for V.
	dM1	Attempts rearrangement using the formula for volume of a sphere to make p^3 the subject Correct order of operations applied to right hand side. Accept arithmetic slips.
	A1	$p = 6.5$ Accept awrt 6.5

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 14 of 33 · Question 11 part b.ii · 5 marks

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Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 15 of 33 · Question 3 part b · 5 marks

2023 JUN · Paper 02 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

3(a)	$675=\frac{\theta r^2}{2}\Rightarrow\theta=\frac{675\times2}{r^2}=\frac{1350}{r^2}$ $P=2r+r\theta\Rightarrow P=2r+r\left(\frac{1350}{r^2}\right)\Rightarrow P=2r+\frac{1350}{r}$	M1 M1A1 cso [3]
(b)	$\frac{dP}{dr}=2-\frac{1350}{r^2}$ $2-\frac{1350}{r^2}=0\Rightarrow r=15\sqrt{3}$ or $\sqrt{675}$ $P=2\times15\sqrt{3}+\frac{1350}{15\sqrt{3}}=60\sqrt{3}$	M1 M1A1 M1A1 [5]
(c)	$\frac{d^2P}{dr^2}=\frac{2700}{r^3}$ $r>0\Rightarrow\frac{d^2P}{dr^2}>0\Rightarrow$ minimum	M1A1ft [2]
		Total 10 marks

Part	Mark	Notes
(a)	**Note:** Accept any variable for P in this part of the question for the first two M marks **only**, even no variable as long as it is clear it is the perimeter.	
	M1	Applies the **correct** formula for the area of a sector with 675 cm^2 and attempts to rearrange to find an expression for θ, r or $r\theta$ Minimally acceptable expression: $\theta=\frac{k}{r^2}, r=\frac{k}{\theta r}$ or $\theta r=\frac{k}{r}$
	M1	Applies the **correct** formula for the perimeter of a sector and substitutes their expression for θ provided it is as a minimum of the form $\theta=\frac{k}{r^2}$ Minimally acceptable expression: $P=2r+\frac{k}{r}$ where k is an integer
	A1	For a fully correct expression for P with no errors seen. You must see $P=$ …….
	ALT – uses the formula $A=\frac{1}{2}rS$ where S is the arc length. S is very popular!	
	M1	Applies the **correct** formula for the area of a sector and rearranges to find an expression for S. $675=\frac{rS}{2}\Rightarrow S=\frac{1350}{r}$ minimally acceptable $S=\frac{k}{r}$
	M1	Applies the correct formula for the perimeter of a sector. $P=2r+S\Rightarrow P=2r+\frac{1350}{r}$ minimally acceptable $P=2r+\frac{k}{r}$
	A1	For a fully correct expression for P with no errors seen. You must see $P=$ …….

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 15 of 33 · Question 3 part b · 5 marks

2023 JUN · Paper 02 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

	ALT 2 – works in degrees	
	M1	Applies a **correct** formula for the area of a sector in degrees and attempts to find an expression for θ $675=\frac{\theta}{360}\times\pi r^2 \Rightarrow \theta=\frac{675\times360}{\pi r^2}=\frac{243000}{\pi r^2}$ Min: $\theta=\frac{K}{\pi r^2}$
	M1	Applies a **correct** formula for the perimeter of a sector and substitutes their expression for θ $P=2r+\frac{\theta}{360}\times2\pi r\left(\frac{243000}{\pi r^2}\right)\Rightarrow P=2r+\frac{1350}{r}$ Min: $P=2r+\frac{K}{r}$ **Do not accept solution with a mix of degrees and radians.**
	A1	For a fully correct expression for P with no errors seen. You must see $P=$ …….
(b)	M1	**NB:** Allow poor notation here, even $\frac{dy}{dx}$ or nothing at all. For attempting to differentiate the **given** expression for P The minimally acceptable expression for the derivate is $\frac{dP}{dr}=2-\frac{Q}{r^2}$ where Q is a **positive** integer.
	M1	For setting their differentiated expression = 0 finding a value for r This is a simple equation to solve. Go through their working checking that it is correct. Do not award this mark for incorrect processing.
	A1	For $r=15\sqrt{3}$ oe [An approximate value is $r=25.98\ldots$.]
	NB: Award the next 2 marks if they appear in part (b) only	
	M1	For substituting their value for r into the given expression for P Only allow this mark if: • They use the correct r and obtain the correct perimeter. • They use an incorrect r provided it is a positive value and **show explicit substitution**
	A1	For the correct final answer in exact form. There is no follow through here.
(c)	**NB: Award the next two marks if they appear in part (c) only.**	
	M1	Finds the second derivative. The minimally acceptable expression for the second derivative is $\frac{d^2P}{dr^2}=\frac{X}{r^3}$ where x is an integer. (The value for $\frac{d^2P}{dr^2}$ is awrt 0.15) If they test $\frac{dP}{dr}$ around the minimum point – send to Review. If they test P either side – score M0A0
	A1ft	For a correct conclusion. FT their 2nd derivative provided it is of the form $\frac{d^2P}{dr^2}=\frac{X}{r^3}$ r must be positive and if they find a value for $\frac{d^2P}{dr^2}$ then substitution **must** be seen unless they use $15\sqrt{3}$ and obtain awrt 0.15

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 15 of 33 · Question 3 part b · 5 marks

2023 JUN · Paper 02 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 16 of 33 · Question 3 part c · 2 marks

2023 JUN · Paper 02 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

3(a)	$675=\frac{\theta r^2}{2}\Rightarrow\theta=\frac{675\times2}{r^2}=\frac{1350}{r^2}$ $P=2r+r\theta\Rightarrow P=2r+r\left(\frac{1350}{r^2}\right)\Rightarrow P=2r+\frac{1350}{r}$	M1 M1A1 cso [3]
(b)	$\frac{dP}{dr}=2-\frac{1350}{r^2}$ $2-\frac{1350}{r^2}=0\Rightarrow r=15\sqrt{3}$ or $\sqrt{675}$ $P=2\times15\sqrt{3}+\frac{1350}{15\sqrt{3}}=60\sqrt{3}$	M1 M1A1 M1A1 [5]
(c)	$\frac{d^2P}{dr^2}=\frac{2700}{r^3}$ $r>0\Rightarrow\frac{d^2P}{dr^2}>0\Rightarrow$ minimum	M1A1ft [2]
		Total 10 marks

Part	Mark	Notes
(a)	**Note:** Accept any variable for P in this part of the question for the first two M marks **only**, even no variable as long as it is clear it is the perimeter.	
	M1	Applies the **correct** formula for the area of a sector with 675 cm^2 and attempts to rearrange to find an expression for θ, r or $r\theta$ Minimally acceptable expression: $\theta=\frac{k}{r^2}, r=\frac{k}{\theta r}$ or $\theta r=\frac{k}{r}$
	M1	Applies the **correct** formula for the perimeter of a sector and substitutes their expression for θ provided it is as a minimum of the form $\theta=\frac{k}{r^2}$ Minimally acceptable expression: $P=2r+\frac{k}{r}$ where k is an integer
	A1	For a fully correct expression for P with no errors seen. You must see $P=$ .....
	ALT – uses the formula $A=\frac{1}{2}rS$ where S is the arc length. S is very popular!	
	M1	Applies the **correct** formula for the area of a sector and rearranges to find an expression for S. $675=\frac{rS}{2}\Rightarrow S=\frac{1350}{r}$ minimally acceptable $S=\frac{k}{r}$
	M1	Applies the correct formula for the perimeter of a sector. $P=2r+S\Rightarrow P=2r+\frac{1350}{r}$ minimally acceptable $P=2r+\frac{k}{r}$
	A1	For a fully correct expression for P with no errors seen. You must see $P=$ .....

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 16 of 33 · Question 3 part c · 2 marks

2023 JUN · Paper 02 (regional R) · extract 2 of 3 · find part c in this mark scheme extract

	ALT 2 – works in degrees	
	M1	Applies a **correct** formula for the area of a sector in degrees and attempts to find an expression for θ $675=\frac{\theta}{360}\times\pi r^2 \Rightarrow \theta=\frac{675\times360}{\pi r^2}=\frac{243000}{\pi r^2}$ Min: $\theta=\frac{K}{\pi r^2}$
	M1	Applies a **correct** formula for the perimeter of a sector and substitutes their expression for θ $P=2r+\frac{\theta}{360}\times2\pi r\left(\frac{243000}{\pi r^2}\right)\Rightarrow P=2r+\frac{1350}{r}$ Min: $P=2r+\frac{K}{r}$ **Do not accept solution with a mix of degrees and radians.**
	A1	For a fully correct expression for P with no errors seen. You must see $P=$ …….
(b)	M1	**NB:** Allow poor notation here, even $\frac{dy}{dx}$ or nothing at all. For attempting to differentiate the **given** expression for P The minimally acceptable expression for the derivate is $\frac{dP}{dr}=2-\frac{Q}{r^2}$ where Q is a **positive** integer.
	M1	For setting their differentiated expression = 0 finding a value for r This is a simple equation to solve. Go through their working checking that it is correct. Do not award this mark for incorrect processing.
	A1	For $r=15\sqrt{3}$ oe [An approximate value is $r=25.98\ldots$.]
	NB: Award the next 2 marks if they appear in part (b) only	
	M1	For substituting their value for r into the given expression for P Only allow this mark if: • They use the correct r and obtain the correct perimeter. • They use an incorrect r provided it is a positive value and **show explicit substitution**
	A1	For the correct final answer in exact form. There is no follow through here.
(c)	**NB: Award the next two marks if they appear in part (c) only.**	
	M1	Finds the second derivative. The minimally acceptable expression for the second derivative is $\frac{d^2P}{dr^2}=\frac{X}{r^3}$ where x is an integer. (The value for $\frac{d^2P}{dr^2}$ is awrt 0.15) If they test $\frac{dP}{dr}$ around the minimum point – send to Review. If they test P either side – score M0A0
	A1ft	For a correct conclusion. FT their 2nd derivative provided it is of the form $\frac{d^2P}{dr^2}=\frac{X}{r^3}$ r must be positive and if they find a value for $\frac{d^2P}{dr^2}$ then substitution **must** be seen unless they use $15\sqrt{3}$ and obtain awrt 0.15

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 16 of 33 · Question 3 part c · 2 marks

2023 JUN · Paper 02 (regional R) · extract 3 of 3 · find part c in this mark scheme extract

Question	Scheme	Marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 17 of 33 · Question 7 part b · 5 marks

2024 NOV · Paper 01 · find part b in this mark scheme extract

7 (a)	$\frac{1}{4}\pi r^2 + 2xr = 100 \quad \left(100 - \frac{1}{4}\pi r^2 = 2xr\right) \Rightarrow x = \frac{50}{r} - \frac{1}{8}\pi r$ oe for example $\frac{100 - \frac{\pi r^2}{4}}{2r}$ $\frac{400 - \pi r^2}{8r}$	M1
	$(P =)\frac{1}{4}(2\pi r) + 4x + 2r$ oe for example $(P =)\frac{1}{2}\pi r + 2x + 2x + 2r$	M1
	$(P =)\frac{1}{2}\pi r + 4\left(\frac{50}{r} - \frac{1}{8}\pi r\right) + 2r \left(= \frac{1}{2}\pi r + \frac{200}{r} - \frac{1}{2}\pi r + 2r\right)$ oe	M1
	$P = \frac{200}{r} + 2r$ *	A1*cso [4]
(b)	$\left(\frac{dP}{dr} =\right) - \frac{200}{r^2} + 2$	M1
	$0 = "-\frac{200}{r^2} + 2" \Rightarrow r =$	M1
	$(r =)10$	A1
	$\left(\frac{d^2P}{dr^2} =\right)\frac{400}{r^3}$	M1
	$r = 10, \frac{d^2P}{dr^2} > 0$ therefore minimum or $\frac{d^2P}{dr^2} = \frac{400}{10^3}$ or $\frac{2}{5} > 0$ therefore minimum	A1 [5]
(c)	$P = \frac{200}{("10")} + 2("10") = 40$	M1 A1 [2]
		Total 11 marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 18 of 33 · Question 8 part b · 5 marks

2022 JUN · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

8 a	Height of the waste paper basket $= \sqrt{(5x)^2 - (3x)^2} = 4x$ (cm)	M1
	$V = \frac{1}{2}(2x+8x)\times 4x \times h = 20x^2h = 2250 \Rightarrow h = \frac{2250}{20x^2}$ oe	M1
	$S = 2\times 20x^2 + 2xh + 2(5xh)$	M1
	$S = 40x^2 + 12x\left(\frac{2250}{20x^2}\right)$ oe	M1
	$S = 40x^2 + \frac{1350}{x}$ *	A1 cso (5)
b	$\frac{dS}{dx} = 80x - \frac{1350}{x^2}$ oe	M1
	$\frac{dS}{dx} = 80x - \frac{1350}{x^2} = 0$ so $x^3 = \frac{135}{8} \Rightarrow x = ...$	M1
	$x = 2.56$ awrt	A1
	$\frac{d^2S}{dx^2} = 80 + \frac{2700}{x^3} > 0$ for all positive values of x ∴ minimum	M1 A1ft (5)
c	When $x = 2.56$ $S = 40(2.56)^2 + \frac{1350}{2.56} = 789$ awrt	M1 A1 (2)
		Total 12 marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 18 of 33 · Question 8 part b · 5 marks

2022 JUN · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	**M1**	For finding the height of the waste-paper basket using Pythagoras theorem. $\sqrt{(5x)^2-(3x)^2}=4x$ (cm)
	M1	For finding the volume which must come from; V = correct area of trapezium (using their height) × h $V=\frac{1}{2}(2x+8x)\times'4x'\times h=2250\Rightarrow('20'x^2h=2250)$ **and** for obtaining an expression for the height h: $h=\frac{2250}{'20'x^2}$ or $xh=\frac{2250}{'20'x}$ Please check their algebra carefully, as some may even substitute $hx^2=...$ **NB: This is an A mark in Epen.**
	M1	For writing the surface area in terms of 2 unknowns [x and h] which need not be simplified. This must be **correct** using their expression in terms of x for the height of the prism. e.g. $S=2\times\frac{(8x+2x)\times'4x'}{2}+2xh+2(5xh)=\left[40x^2+12xh\right]$
	M1	For eliminating h from their expression for S S must be in the form Ax^2+Bxh and h must be of the form $\frac{C}{x^2}$ where A, B and C are constants.
	A1 cso	For the given result exactly as written. $S=40x^2+\frac{1350}{x}$
(b)	**M1**	For an acceptable attempt to differentiate the **given** expression for S. See General Guidance.
	M1	For setting their $\frac{dS}{dx}=0$ and attempting to find a value for x The minimum acceptable expression is $\frac{dS}{dx}=Px\pm\frac{Q}{x^2}$ where P and Q are constants.
	A1	For awrt $x=2.56$
	M1	For differentiating their $\frac{dS}{dx}$, which must be as a minimum $\frac{dS}{dx}=Px\pm\frac{Q}{x^2}$ to find the second derivative to achieve as a minimum $\frac{d^2S}{dx^2}=\pm M\pm\frac{N}{x^3}$
	A1ft	Concludes that as $\frac{d^2S}{dx^2}$ will always be positive, [either by substitution, or by inference] so the value of x obtained will be a minimum. $\left[\frac{d^2S}{dx^2}=80+\frac{2700}{2.56^3}=240.9...\right]$ with a conclusion. **NOTE:** The ft only applies to their value of x. Do not ft an incorrect $\frac{d^2S}{dx^2}$
(c)	**M1**	Substitutes their value of x, obtained using a correct method) into the **given** expression for S [provided it is a positive value, do **NOT** allow negative values of x]
	A1	For awrt 789

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 18 of 33 · Question 8 part b · 5 marks

2022 JUN · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 19 of 33 · Question 7 part b · 4 marks

2022 JAN · Paper 01 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

7 (a)	(Surface area =) $12x^2 + 4xy + 3xy + 5xy = 144$ oe	M1
	$12x^2 + 12xy = 144$ oe leading to $\frac{144 - 12x^2}{12x}$ oe	A1
	$V = \frac{1}{2} \times 3x \times 4x \times y \ (= 6x^2 y)$	M1
	$= 6x^2 \times \frac{144 - 12x^2}{12x} = 6x(12 - x^2)$	M1
	$= 72x - 6x^3$ *	A1 cso [5]
(b)	$\left(\frac{dV}{dx}\right) = 72 - 18x^2$	M1
	$\left(\frac{dV}{dx} = \right)"72 - 18x^2" = 0$ $\Rightarrow x =$	M1 (A1 on ePen)
	$x = 2$	A1 (M1 on ePen)
	$\left(\frac{d^2V}{dx^2}\right) = -36x$	B1 (A1 on ePen)
	$x = 2 \ \frac{d^2V}{dx^2} < 0$ Therefore maximum	[4]
(c)	$V = 72 \times 2 - 6 \times 2^3 = 96$	M1 A1 [2]
		Total 11 marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 19 of 33 · Question 7 part b · 4 marks

2022 JAN · Paper 01 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	M1	For surface area as an equation in x and y (correct and unsimplified)
	A1	For $12x^2+12xy=144$ oe and correctly making y the subject $\frac{144-12x^2}{12x}$ (oe)
	M1	$V=\frac{1}{2}\times 3x\times 4x\times y$ $(=6x^2y)$
	M1	For correct substitution of their expression for y, which must be in terms of x only, into their formula for V.
	A1*cso	Obtains the given expression
(b)	M1	For attempting to differentiate V wrt x At least one term must be fully correct, the other – follow general guidance
	M1 (A1 on ePen)	For setting their derivative (which must involve a changed expression) = 0 and a complete solution to find the value of x
	A1 (M1 on ePen)	For $x = 2$ only
	B1 (A1 on ePen)	For a correct second derivative and correct justification that this is a maximum, with some form of conclusion. All work to be correct for this mark. i.e it is acceptable, as this is a simple substitution to state if $x = 2$, $\frac{d^2V}{dx^2}<0$, but if a substitution of $x = 2$ is made, the value for the second derivative must be given as – 72.
ALT final B1	B1 (A1 on ePen)	Testing and substituting into the correct second derivative with appropriate values either side of $x = 2$. For correct justification that this is a maximum, with some form of conclusion. All work to be correct for this mark
(c)	M1	For correct substitution of their x into V
	A1	96

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 19 of 33 · Question 7 part b · 4 marks

2022 JAN · Paper 01 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 20 of 33 · Question 9 part c · 1 mark

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

Number		
9 (a)	$540 = 3x^2h \Rightarrow h = \frac{180}{x^2}$ $S = 2\left(3x^2 + 3xh + xh\right) = 6x^2 + 8xh$ $\Rightarrow S = 6x^2 + 8x \times \frac{180}{x^2} = 6x^2 + \frac{1440}{x}$ *	M1 M1 depM1A1 [4]
(b)	$S = 6x^2 + 1440x^{-1}$ $\frac{dS}{dx} = 12x - 1440x^{-2}$ At min/max $\frac{dS}{dx} = 0$ $12x - 1440x^{-2} = 0 \Rightarrow x^3 = 120 \Rightarrow x = 4.93242....$ $x \approx 4.93$ (3sf) $\frac{d^2S}{dx^2} = 12 + \frac{2880}{x^3} \Rightarrow$ Always positive for positive values of x, hence minimum	M1 M1A1 M1A1ft [5]
(c)	$S = 6 \times 4.93242^2 + \frac{1440}{4.93242} = 437.9185 \approx 438$	B1 [1]
		Total 10 marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 20 of 33 · Question 9 part c · 1 mark

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 21 of 33 · Question 8 part c · 2 marks

2022 JAN · Paper 01 · extract 1 of 3 · find part c in this mark scheme extract

8 (a)	$90\pi = \pi r^2 h \Rightarrow h = \dfrac{90}{r^2}$ $S = 2\pi r^2 + 2\pi rh \Rightarrow S = 2\pi r^2 + 2\pi r \times \dfrac{90}{r^2}$ $S = 2\pi r^2 + \dfrac{2\times 90\pi}{r} = 2\pi r^2 + \dfrac{180\pi}{r}$ *	B1 M1 A1cso [3]
(b)	$\dfrac{dS}{dr} = 4\pi r - \dfrac{180\pi}{r^2}$ $\dfrac{dS}{dr} = 0 \Rightarrow 4\pi r - \dfrac{180\pi}{r^2} = 0 \Rightarrow 4\pi r = \dfrac{180\pi}{r^2} \Rightarrow r^3 = 45 \Rightarrow r = \ldots$ $r = 3.55689\ldots \Rightarrow r \approx 3.56$ $\dfrac{d^2S}{dr^2} = 4\pi + \dfrac{360\pi}{r^3}$ $\dfrac{d^2S}{dr^2} = 4\pi + \dfrac{360\pi}{r^3} \Rightarrow \left(\dfrac{d^2S}{dr^2} = 37.699\ldots\right)$ $37.699 > 0 \Rightarrow$ hence minimum	M1 M1 A1 M1 A1ft [5]
(c)	$S = 2\pi \times 3.556\ldots^2 + \dfrac{2\times 90\pi}{3.556..} = \ldots$ $S = 238.4769\ldots \Rightarrow S = 238\ \left(\text{cm}^2\right)$	M1 A1 [2]
		Total 10 marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 21 of 33 · Question 8 part c · 2 marks

2022 JAN · Paper 01 · extract 2 of 3 · find part c in this mark scheme extract

		$S = 2\pi r^2 + \frac{2\times 90\pi}{r} = 2\pi r^2 + \frac{180\pi}{r}$ Must have the $S =$ for this mark.
(b)	**M1**	For attempting to differentiate the given expression for S at least one power to decrease and neither power to increase. $\frac{dS}{dr} = 4\pi r - \frac{180\pi}{r^2}$
	M1	Sets their $\frac{dS}{dr} = 0$ and attempts to solve for r $4\pi r - \frac{180\pi}{r^2} = 0 \Rightarrow 4\pi r = \frac{180\pi}{r^2} \Rightarrow r^3 = 45 \Rightarrow r = ...$
	A1	For the correct value of $r = 3.55689... \Rightarrow r \approx 3.56$ Accept awrt 3.56
	M1	For attempting to differentiate their expression for $\frac{dS}{dr}$ at least one power to decrease and neither power to increase. $\frac{d^2S}{dr^2} = 4\pi + \frac{360\pi}{r^3}$
	A1ft	For correct work throughout $\frac{d^2S}{dr^2} = 4\pi + \frac{180\pi}{r^3} \Rightarrow \left(\frac{d^2S}{dr^2} = 37.699...\right)$ $37.699 > 0 \Rightarrow$ hence minimum Evaluation not required as both terms positive so $\frac{d^2S}{dr^2} > 0$ hence minimum Indication of positive or >0 required. If $\frac{d^2S}{dr^2}$ evaluated incorrectly then do not award. If evaluated then accept awrt 38
(c)	**M1**	For substituting their value of r into the **given** expression for S $S = 2\pi \times '3.556'...^2 + \frac{2\times 90\pi}{'3.556'..} = ...$ Their value of r >0
	A1	$S = 238.4769... \Rightarrow S = 238\ (\text{cm}^2)$ Accept awrt 238

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 21 of 33 · Question 8 part c · 2 marks

2022 JAN · Paper 01 · extract 3 of 3 · find part c in this mark scheme extract

Question	Scheme	Marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 22 of 33 · Question 8 part c · 2 marks

2023 JAN · Paper 01 · extract 1 of 3 · find part c in this mark scheme extract

8 a	(Area) $= 2xy + \frac{1}{2}x^2\left(\frac{1}{2}\right)\left(= 2xy + \frac{1}{4}x^2 = 50\right)$	M1
	So $y = \frac{50 - \frac{1}{4}x^2}{2x}\left(= \frac{25}{x} - \frac{1}{8}x\right)$	dM1
	$P = 2x + 4y + \frac{1}{2}x = \frac{5}{2}x + 4y$	M1
	$= \frac{5}{2}x + 4\left(\frac{25}{x} - \frac{1}{8}x\right) \Rightarrow P = 2x + \frac{100}{x}$ *	M1 A1 cso (5)
b	$\left(\frac{dP}{dx} =\right) 2 - \frac{100}{x^2}$	M1 A1
	$\left(\frac{dP}{dx} = 0 \Rightarrow\right) 2 - \frac{100}{x^2} = 0 \Rightarrow x^2 = 50 \Rightarrow x = 5\sqrt{2}$ oe	M1 A1
	$\frac{d^2P}{dx^2} = \frac{200}{x^3}$ When $x = 5\sqrt{2}$ $\frac{d^2P}{dx^2} = \frac{2}{5}\sqrt{2}(= 0.565.....) > 0$ therefore minimum	M1 A1 (6)
c	$P = 2\left(5\sqrt{2}\right) + \frac{100}{5\sqrt{2}} = 20\sqrt{2}$	M1 A1 (2)
		Total 13 marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 22 of 33 · Question 8 part c · 2 marks

2023 JAN · Paper 01 · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For writing an expression (may not be simplified) for the area in terms of x and y. Allow any expression of the form $xy + xy + ax^2$ oe where $a > 0$ This mark can be implied by an equation for A with the required $xy + xy + ax^2$ oe where $a > 0$
	dM1	For placing their expression for area = 50 and correctly rearranging to make y the subject. The expression for y does not need to be simplified.
	M1	For correctly writing an equation for perimeter in terms of x and y (may not be simplified). There must be a $P =$, though this may be seen later in their work. As long as $P =$ appears by the end of the work, this mark can be awarded. Perimeter = instead of $P =$ is acceptable for this mark
	M1	For correct substitution of their expression for y into their expression for **P or the Perimeter to reach an expression for P which must be in terms of x only. Simplification not required.**
	A1*cso	Obtains the given result with no errors. Perimeter = instead of $P =$ is **NOT** acceptable for this mark
For this question, if we are confident any second solution provided is a re-start and the work the candidate wishes us to consider, this is the work can be marked.		
(b)	M1	For an attempt to differentiate the given expression for P wrt x It is not necessary to see $\frac{dP}{dx} =$ Award this mark for any expression of the form $2 \pm \frac{b}{x^2}$, $b \neq 0$ oe
	A1	For $2 - \frac{100}{x^2}$ oe
	M1	For placing their expression for $\frac{dP}{dx} = 0$ which must have a term in x^2, a correct rearrangement to solve the equation formed and an attempt to find a value for x
	A1	For $x = 5\sqrt{2}$ oe ignore $x = -5\sqrt{2}$
	M1	For $\frac{d^2P}{dx^2} = \pm\frac{c}{x^3}$ $c \neq 0$
	A1	For substituting in a correct value for x, evaluating correctly, stating > 0 and drawing a conclusion. This can be as simple as # or shown. Students may also argue that the second derivative has to be positive as x is positive. This must be a convincing argument that since x is positive the derivative is positive and then a valid minimal conclusion. Do not award this mark if a negative value of x is used and it is not discounted at some point.
	ALT M1	Final 2 marks For substitution of valid values for x either side of their value of x found for the minimum value into $\frac{dP}{dx} = 2 \pm \frac{b}{x^2}$, $b \neq 0$
	A1	For fully correct substitution of their values for x into a correct expression for a first derivative, correctly evaluating or arguing the sign of the derivative and a conclusion drawn. Do not award this mark if a negative value of x is used and it is not discounted at some point.
(c)	M1	For substitution of their value for $x = 5\sqrt{2}$ oe into the given expression for P
	A1	For $20\sqrt{2}$

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 22 of 33 · Question 8 part c · 2 marks

2023 JAN · Paper 01 · extract 3 of 3 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 23 of 33 · Question 7 part c · 2 marks

2023 NOV · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

7(a)	$S = 5x^3 + (3x-4)^2$ $\Rightarrow S = 5x^3 + 9x^2 - 24x + 16$ *	M1 A1 cso [2]
(b)	$\frac{dS}{dx} = 15x^2 + 18x - 24 = 0$ $\Rightarrow (5x-4)(x+2) = 0 \Rightarrow x = \frac{4}{5}, -2$ $\frac{d^2S}{dx^2} = 30x + 18 = 30\left(\frac{4}{5}\right) + 18 \Rightarrow$ +ve hence minimum	M1 M1A1 M1A1 [5]
(c)	$S = 5\left(\frac{4}{5}\right)^3 + 9\left(\frac{4}{5}\right)^2 - 24\left(\frac{4}{5}\right) + 16 = \frac{128}{25}$ or 5.12	M1A1 [2]
		Total 9 marks

Part	Mark	Notes
(a)	M1	For substituting y into the given S Substituting x will not yield the required expression.
	A1 cso	For obtaining the given expression with no errors. You must check every line of their working.
(b)	M1	For an attempt to differentiate **the given** expression for S wrt x, Accept **at least** two terms fully correct with no power of x to increase.
	M1	Sets their differentiated expression = 0 and attempts to solve, **provided** it is a quadratic. See General Guidance for the definition of an attempt to solve a QE
	A1	For the correct **two** values of x.
	M1	Attempts to differentiate again. Minimally acceptable attempt is $\left(\frac{d^2S}{dx^2}\right) = Ax + B$
	A1	Conclusion: Concludes that the positive value of x $\left(\frac{4}{5}\right)$ will give a positive $\frac{d^2S}{dx^2}$ hence will be a minimum. For example, positive + positive = positive hence minimum. OR Substitutes either value of x, with the appropriate conclusion and correctly concludes that $x = \frac{4}{5}$ gives a minimum.

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 23 of 33 · Question 7 part c · 2 marks

2023 NOV · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

		NOTE: If they evaluate $\frac{d^2S}{dx^2}$ it must be correct [= 42] An incorrect value is A0.
(c)	M1	Uses **their** minimum value of x in the **given** expression, even if they identify $x = -2$ as the value giving a minimum S. If their minimum is not explicitly identified, then award M0. If they substitute both values without identifying which is which, award M0.
	A1	For the correct value of S

Question	Scheme	Marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 24 of 33 · Question 11 part c · 1 mark

2023 JAN · Paper 02 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

11	$S = \pi r^2 + 2\pi rh$ $\pi r^2 + 2\pi rh = 625\pi \Rightarrow 2rh = 625 - r^2 \Rightarrow h = \dfrac{625 - r^2}{2r}$ *	B1 M1A1 [cso] [3]
(b)	$V = \pi r^2\left(\dfrac{625 - r^2}{2r}\right)$ or $\left(V = \dfrac{625\pi}{2}r - \dfrac{\pi r^3}{2}\right)$ $\dfrac{dV}{dr} = \dfrac{625\pi}{2} - \dfrac{3\pi r^2}{2}$ $\dfrac{625\pi}{2} - \dfrac{3\pi r^2}{2} = 0 \Rightarrow 625 = 3r^2 \Rightarrow r = 14.4337... \Rightarrow r \approx 14.4$ For differentiating their $\dfrac{dV}{dr}$ $\dfrac{d^2V}{dr^2} = -3\pi r \ (= -136.03\ ...)$ For concluding that as $\dfrac{d^2V}{dr^2}$ is negative then the volume is a maximum. (Explicit substitution of r is not required).	M1 M1 M1A1 M1 A1 [6]
(c)	$h = \dfrac{625 - 14.43375^2}{2 \times 14.43375} = 14.4337... \approx 14.4$	B1ft [1]
		Total 10 marks

Question	Scheme	Marks
11	For a correct expression for the surface area of the cup. $S = \pi r^2 + 2\pi rh$	B1
	For equating their expression for the surface area to 625π and attempting to make h the subject. $\pi r^2 + 2\pi rh = 625\pi \Rightarrow 2rh = 625 - r^2 \Rightarrow h = \dfrac{625 - r^2}{2r}$ Allow errors in arithmetic but not mathematically incorrect process.	M1
	For the correct expression for h as shown with no errors. $h = \dfrac{625 - r^2}{2r}$*	A1 cso [3]

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 24 of 33 · Question 11 part c · 1 mark

2023 JAN · Paper 02 (regional R) · extract 2 of 3 · find part c in this mark scheme extract

(b)	For the correct volume of the container in terms of r only. $V = \pi r^2\left(\frac{625-r^2}{2r}\right)$ or $\left(V = \frac{625\pi}{2}r - \frac{\pi r^3}{2}\right)$	M1
	Minimally acceptable attempt at differentiation, see general guidance, no power to increase. $\frac{dV}{dr} = \frac{625\pi}{2} - \frac{3\pi r^2}{2}$	M1
	Places their derivative = 0 and attempts to rearrange to find r. Minimally acceptable derivative is of the form $a \pm b\pi r^2$ $\frac{625\pi}{2} - \frac{3\pi r^2}{2} = 0 \Rightarrow 625 = 3r^2 \Rightarrow r = ...$	M1
	For the correct value of r $r = 14.4337... \Rightarrow r \approx 14.4$ (cm) Must reject negative value if found, award A0 if not rejected. May be implied from subsequent working.	A1
	For differentiating their $\frac{dV}{dr}$ For minimally acceptable attempt to differentiate their first derivative, see general guidance, no power to increase. Or testing gradients or a sketch. $\frac{d^2V}{dr^2} = -3\pi r$	M1
	For concluding that as $\frac{d^2V}{dr^2}$ is negative then the volume is a maximum. (Explicit substitution of r is not required).	A1 [6]
(c)	For the value of h using their value of r $h = \frac{625-14.43375^2}{2\times 14.43375} = 14.4337... \approx 14.4$ Allow for 14.5 if substituting $r = 14.4$	B1ft [1]
		Total 10 marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 24 of 33 · Question 11 part c · 1 mark

2023 JAN · Paper 02 (regional R) · extract 3 of 3 · find part c in this mark scheme extract

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Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 25 of 33 · Question 6 part c · 1 mark

2025 JUN · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

6 (a)	$700\pi = 2\pi r^2 + 2\pi rh$	B1
	$h = \dfrac{350 - r^2}{r}$ or $rh = 350 - r^2$	M1
	$V = \pi r^2 h = \pi r^2\left(\dfrac{350 - r^2}{r}\right) = \left\{\pi r\left(350 - r^2\right)\right\} = \pi r\left(350 - r^2\right)$*	M1A1* cso [4]
(b)	$V = 350\pi r - \pi r^3$	
	$\dfrac{\mathrm{d}V}{\mathrm{d}r} = 350\pi - 3\pi r^2 = 0$	M1
	$\Rightarrow 350\pi = 3\pi r^2 \Rightarrow r = \sqrt{\dfrac{350}{3}} = 10.8$ (3sf)	M1A1
	$\dfrac{\mathrm{d}^2V}{\mathrm{d}r^2} = -6\pi r$	M1
	$\dfrac{\mathrm{d}^2V}{\mathrm{d}r^2} < 0 \therefore$ max	A1 [5]
(c)	$h = \dfrac{350 - r^2}{r} = \dfrac{350 - \dfrac{350}{3}}{\sqrt{\dfrac{350}{3}}} = 21.6$ (3 s.f.)	B1 [1]
		Total 10 marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 25 of 33 · Question 6 part c · 1 mark

2025 JUN · Paper 02 · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Additional Guidance
(a)	B1	For a **correct** equation for the surface area in terms of the radius and height
	M1	For rearranging to find h in terms of r which must be as a minimum $h = \frac{k - lr^2}{mr}$ where k, l and m are integers OR $rh = k - lr^2$ where k and l are integers
	M1	For a correct expression for volume using *their* h in terms of r
	A1* cso	For rearranging and obtaining the given result with no errors or omissions and must include $V = \ldots$ [Allow Volume = …. or Vol = …..]
(b)	M1	For an attempt to differentiate the **given** expression for V which must be of the form $\frac{dV}{dr} = 350\pi - K\pi r^2$ where K is a constant Using Product Rule: $\frac{dV}{dr} = \pi r(-2r) + \pi(350 - r^2) \Rightarrow \frac{dV}{dr} = \left[350\pi - K\pi r^2\right]$
	M1	For setting their $\frac{dV}{dr} = 0$ **and** attempting to find a value for r.
	A1	$r = \sqrt{\frac{350}{3}}$ [=10.801...] accept awrt 10.8
	M1	For finding $\frac{d^2V}{dr^2}$ Accept only $\frac{d^2V}{dr^2} = \pm k\pi r$ where k is a constant See general guidance for what constitutes an attempt to differentiate.
	A1	For the correct conclusion derived from the correct working throughout.
(c)	B1	For obtaining the value of h correct to 3 s.f. Accept awrt 21.6

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 25 of 33 · Question 6 part c · 1 mark

2025 JUN · Paper 02 · extract 3 of 3 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 26 of 33 · Question 11 part c · 2 marks

2020 JAN · Paper 01 · extract 1 of 5 · find part c in this mark scheme extract

11 (a)	Let x = the length of the side of the triangle and h = the length of the prism	
	$\frac{1}{2}x^2 \sin 60\, h = 72$ or $\frac{1}{2}\left(\sqrt{\left(x^2 - \left(\frac{1}{2}x\right)^2\right)}\right)xh = 72$	M1
	$\frac{\sqrt{3}x^2h}{4} = 72$	M1
	$h = \frac{288}{\sqrt{3}x^2}$	A1
	$S = 2 \times \frac{1}{2}x^2 \sin 60 + 3xh$ or $2 \times \frac{1}{2}\left(\sqrt{\left(x^2 - \left(\frac{1}{2}x\right)^2\right)}\right)x + 3xh$	M1
	$S = \frac{\sqrt{3}x^2}{2} + 3x\left(\frac{288}{\sqrt{3}x^2}\right)$	M1
	$S = \frac{\sqrt{3}x^2}{2} + \frac{288\sqrt{3}}{x}$ *	A1 cso (6)
(b)	$\frac{dS}{dx} = \sqrt{3}x - \frac{288\sqrt{3}}{x^2} (= 0)$	M1
	$x^3 = 288$	dM1
	$x = \sqrt[3]{288} = 6.604$	A1
	$\frac{d^2S}{dx^2} = \sqrt{3} + \frac{576\sqrt{3}}{x^3}$	ddM1
	$\frac{d^2S}{dx^2} > 0$ (when x =6.6) ∴ value is a minimum	A1 (5)
(c)	Substitutes their x into $S = \frac{\sqrt{3}x^2}{2} + \frac{288\sqrt{3}}{x}$	M1
	$S = 113$	A1 (2)
		[13]

Part	Mark	Additional Guidance
(a)	M1	For the correct expression for the volume of the prism in terms of x and h (or other letter for the length, e.g. l) Simplification not required for this mark $72 = \left(\frac{1}{2} \times x \times x \times \sin 60^\circ\right) \times h$ or $72 = \left(\frac{1}{2} \times x \times \sqrt{x^2 - \frac{x^2}{4}}\right) \times h$ or $72 = \left(\frac{\sqrt{3}}{4}x^2\right) \times h$
	M1	For an attempt to find an expression for h in terms of x Accept as a minimum $h = \frac{k}{x^2}$ where k is a positive integer
	A1	For $h = \frac{288}{\left(\sqrt{3}\right)x^2}$ or $h = \frac{96\sqrt{3}}{x^2}$

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 26 of 33 · Question 11 part c · 2 marks

2020 JAN · Paper 01 · extract 2 of 5 · find part c in this mark scheme extract

	M1	For an expression for S in terms of x and h (ft their area of the triangle) $S=2\left(\frac{1}{2}\times x\times x\times\sin 60^\circ\right)+3xh$ or $S=2\left(\frac{1}{2}\times x\times\sqrt{x^2-\frac{x^2}{4}}\right)+3xh$ $\left(S=\frac{\sqrt{3}}{2}x+3xh\right)$
	M1	For substituting their h into their S $S=2\left(\frac{1}{2}\times x\times x\times\sin 60^\circ\right)+\left(3x\times\frac{288}{\left(\sqrt{3}\right)x^2}\right)$ or $S=2\left(\frac{1}{2}\times x\times\sqrt{x^2-\frac{x^2}{4}}\right)+\left(3x\times\frac{288}{\left(\sqrt{3}\right)x^2}\right)$
	A1	For the correct expression for S as given. **The expression must be set equal to S.** $S=\frac{\sqrt{3}x^2}{2}+\frac{288\sqrt{3}}{x}$ exactly as seen here. * This is a given result so full working must be seen.
(b)	M1	For an attempt to differentiate the **given** expression for S $\frac{dS}{dx}=\sqrt{3}x-\frac{288\sqrt{3}}{x^2}$ or $\frac{dS}{dx}=\sqrt{3}x-288\sqrt{3}x^{-2}$ (See General Guidance for the definition of an attempt)
	dM1	For setting their differentiated expression = 0 and attempting to solve for x $\sqrt{3}x-\frac{288\sqrt{3}}{x^2}=0\Rightarrow x^3=288\Rightarrow x=\left(6.604\right)$ (rounded correctly) This mark is dependent on the first M mark in (b)
	A1	For $x = 6.604$ rounded correctly
	dM1	For attempting the second derivative (usual definition of an attempt) $\frac{d^2S}{dx^2}=\sqrt{3}+\frac{576\sqrt{3}}{x^3}$ This mark is dependent on first M mark in (b)
	A1ft	Concludes either that $\frac{d^2S}{dx^2}>0$ for all **positive** values of x **or** substitutes in their value of x to show that $\frac{d^2S}{dx^2}=5.19.....$ hence positive so must be a minimum. Only ft if the final conclusion is a minimum provided their $\frac{d^2S}{dx^2}$ is algebraically correct
	ALT – for justifying the minimum using their derivative	
	dM1	Chooses a value either side of their value of x and substituting them into their $\frac{dS}{dx}$ e.g. $x = 6$ and 7 $\frac{dS}{dx}=\sqrt{3}\times 6-\frac{288\sqrt{3}}{6^2}=-3.46...$ and $\frac{dS}{dx}=\sqrt{3}\times 7-\frac{288\sqrt{3}}{7^2}=1.944...$
	A1ft	Concludes that the gradient function moves from negative to positive hence must be a minimum.
(c)	M1	Substitutes their value of x into the given expression for S $S=\frac{\sqrt{3}\text{'}6.604\text{'}^2}{2}+\frac{288\sqrt{3}}{\text{'}6.604\text{'}}=...$
	A1	For $S = 113$ rounded correctly

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 26 of 33 · Question 11 part c · 2 marks

2020 JAN · Paper 01 · extract 3 of 5 · find part c in this mark scheme extract

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 26 of 33 · Question 11 part c · 2 marks

2020 JAN · Paper 01 · extract 4 of 5 · find part c in this mark scheme extract

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 26 of 33 · Question 11 part c · 2 marks

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Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 27 of 33 · Question 5 part c · 2 marks

2023 JUN · Paper 01 · extract 1 of 3 · find part c in this mark scheme extract

5(a)	$V = x \times 4x \times h \Rightarrow 4x^2h = 75 \Rightarrow h = \frac{75}{4x^2}$ $S = 2\left(4x^2 + xh + 4xh\right) = \left[8x^2 + 2xh + 8xh\right] \Rightarrow S = 8x^2 + 10xh$ $\Rightarrow S = 8x^2 + 10x \times \frac{75}{4x^2} \Rightarrow S = 8x^2 + \frac{375}{2x}$ *	B1 M1 M1A1 cso [4]
(b)(i)	$\frac{dS}{dx} = 16x - \frac{375}{2x^2} = 0$ $\Rightarrow 16x = \frac{375}{2x^2} \Rightarrow x^3 = \frac{375}{32} \Rightarrow x = 2.2714... \approx 2.27 \ (\text{cm})$	M1 M1A1
(ii)	$\frac{d^2S}{dx^2} = 16 + 375x^{-3} \approx 48$ Conclusion: (As x is always positive) $\frac{d^2S}{dx^2} > 0 \Rightarrow$ minimum	M1 A1ft [5]
(c)	$S = 8 \times '2.27'^2 + \frac{375}{2 \times '2.27'} = 123.822... \approx 124 \ (\text{cm}^2)$	M1A1 [2]
		Total 11 marks

Part	Mark	Notes
(a)	**B1**	For finding the correct expression for h in terms of x Accept this mark seen anywhere in part (a) Accept also for example, $xh = \frac{75}{4x}$ or any equivalent that can be substituted into an expression for S.
	M1	For finding an expression for S is terms of x and h. Accept as a minimum $Ax^2 + Bxh$ where A and B are constants. Accept other letters for S at this stage. E.g SA etc. or even no letter at all.
	M1	For substituting their h into their S
	A1	For the correct expression for S as shown in the question. You must see $S = 8x^2 + \frac{375}{2x}$ including S although S seen at the top of a column of working is also fine.
(b)	**M1**	For an attempt to differentiate the given S which as a minimum must be of the form

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 27 of 33 · Question 5 part c · 2 marks

2023 JUN · Paper 01 · extract 2 of 3 · find part c in this mark scheme extract

		$\frac{dS}{dx} = 16x - \frac{k}{x^2}$ where k is a constant.
	M1	For setting their $\frac{dS}{dx} = 0$ and attempting to solve the equation and reaching a value of x. It must be a clearly differentiated expression even if they do not score the first M mark in (b). That is, they must reduce the power of at least one term by 1.
	A1	For awrt $x = 2.27$
	M1	For an attempt to differentiate their $\frac{dS}{dx}$ to find $\frac{d^2S}{dx^2}$ which must be as a minimum $\frac{d^2S}{dx^2} = 16 + lx^{-3}$ where l is a constant **Other methods** Please send to Review any examples of candidates who test the gradient on either side of the minimum, or who draw a sketch.
	A1ft	Concludes that as $x > 0$ so $\frac{d^2S}{dx^2} > 0$ and therefore S is a minimum when $x = 2.27$ Dependent on correct method seen. Follow through their value of x provided it is positive. **NB:** They do not need to evaluate $\frac{d^2S}{dx^2}$ to score this mark. [See above] However, if they use the correct [2.27] or an incorrect positive value of x, provided substitution is seen award the mark even if the final evaluation is incorrect. If no substitution is seen and either the value of x is incorrect or they do not obtain approximately 48, withhold this mark.
(c)	**M1**	Substitutes their $x = 2.27$ into the given S Follow through their value of x provided it is a positive value. Use of a negative value of x is M0. If the final value of S is incorrect following an incorrect value of x award this mark only for explicit substitution seen.
	A1	For awrt $S = 124$ (cm^2)

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 27 of 33 · Question 5 part c · 2 marks

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Question	Scheme	Marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 28 of 33 · Question 8 part c · 2 marks

2024 JUN · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

8(a)	Finds the length DE $x^2 = (DE)^2 + (DE)^2 \Rightarrow DE = \frac{x}{\sqrt{2}}$ or $DE^2 = \frac{x^2}{2}$ Volume of prism $3.6 = \left(\frac{x}{\sqrt{2}}\right)^2 \times \frac{1}{2} \times y \Rightarrow \left[y = \frac{72}{5x^2}\right]$ Surface area of prism $S = 2 \times \frac{x^2}{4} + 2 \times \frac{x}{\sqrt{2}} \times \frac{72}{5x^2} + x \times \frac{72}{5x^2} = \left[\frac{x^2}{2} + \sqrt{2} \times \frac{72}{5x} + \frac{72}{5x}\right]$ $\Rightarrow S = \frac{x^2}{2} + \frac{72(\sqrt{2}+1)}{5x}$ *	B1 M1 M1 A1 cso [4]
(b)	$\frac{dS}{dx} = x - \frac{72(\sqrt{2}+1)}{5x^2} = 0$ $\Rightarrow \frac{72(\sqrt{2}+1)}{5x^2} = x$ $\Rightarrow x = \sqrt[3]{\frac{72(\sqrt{2}+1)}{5}} = 3.26371\ldots \approx 3.26\ (\text{cm})$ $\frac{d^2S}{dr^2} = 1 + \frac{144(\sqrt{2}+1)}{5x^3}$ $\Rightarrow$ positive + positive = positive $\Rightarrow$ is always greater than 0, hence minimum	M1 M1 A1 B1ft [4]
(c)	$S_{(\min)} = \frac{3.26371^2}{2} + \frac{72(\sqrt{2}+1)}{5 \times 3.26371} = 15.97779\ldots \approx 16\ (\text{cm}^2)$	M1A1 [2]
		Total 10 marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 28 of 33 · Question 8 part c · 2 marks

2024 JUN · Paper 02 · extract 2 of 3 · find part c in this mark scheme extract

(a)	B1	For using Pythagoras theorem or any appropriate trigonometry to find the length DE (or any equivalent side)
	M1	For using the correct formula to find the volume of a prism and rearranging to find an expression for y minimally of the form $y = \frac{A}{Bx^2}$ This need not be simplified.
	M1	For forming an expression for S using their y The basic expression for the surface area must be correct, with correct substitution of their values. $S = 2\times'DE'+2\times'DE'\times y+xy \Rightarrow S = 2\times'\frac{x^2}{4}'+2\times'\frac{x}{\sqrt{2}}'\times\left('\frac{72}{5x^2}'\right)+x\times\left('\frac{72}{5x^2}'\right)$
	A1 cso	For $S = \frac{x^2}{2}+\frac{72\left(\sqrt{2}+1\right)}{5x}$ as written
(b)	M1	For differentiating the **given** expression for S Minimally acceptable differentiated expression is as follows: $\frac{dS}{dx} = x \pm Kx^{-2}$ oe. where K is a constant
	M1	For setting their $\frac{dS}{dx} = 0$ and attempting to solve to obtain a value for x
	A1	For the correct value of x Accept awrt 3.26 (cm)
	B1ft	For differentiating their $\frac{dS}{dr}$ to minimally obtain $\frac{d^2S}{dr^2} = 1 \pm Lx^{-3}$ and coming to an appropriate conclusion. The justification has to be correct and complete. If they refer/use a substitution it must be seen and it must be correct.. That is, $\frac{d^2S}{dr^2} = 1+\frac{144\left(\sqrt{2}+1\right)}{5\times3.26^3} = 3.00...$ $3 > 0$ so positive, hence minimum Alternatively: x is positive so; positive $+ \frac{\text{positive}}{\text{positive}} \Rightarrow$ positive hence minimum
(c)	M1	For substituting their value of x (provided it is positive) and attempting to find a value of S
	A1	For awrt $S = 16$

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 28 of 33 · Question 8 part c · 2 marks

2024 JUN · Paper 02 · extract 3 of 3 · find part c in this mark scheme extract

Question	Scheme	Marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 29 of 33 · Question 5 part c · 2 marks

2025 NOV · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

5(a)	$\left[S = 2\pi rh + 2\pi r^2 \Rightarrow 322\pi = 2\pi rh + 2\pi r^2\right]$ $\Rightarrow (h =)\dfrac{161 - r^2}{r}$ oe eg $\dfrac{161}{r} - r$ eg $\dfrac{322\pi - 2\pi r^2}{2\pi r}$ $\left(V = \pi r^2 h \Rightarrow\right)(V =)\pi r^2\left("\dfrac{161 - r^2}{r}"\right) \Rightarrow$ $V = \pi r\left(161 - r^2\right)$ *	B1 M1 A1cso [3]
ALT	$\left(S = 2\pi rh + 2\pi r^2 \Rightarrow 322\pi = 2\pi rh + 2\pi r^2 \Rightarrow 161 = rh + r^2\right)$ $\Rightarrow (rh =) 161 - r^2$ $\left(V = \pi r^2 h \Rightarrow\right)(V =)\pi r\left(161 - r^2\right)$ *	B1 M1 A1
(b)	$\left[\dfrac{dV}{dr} =\right] 161\pi - 3\pi r^2$ oe $161\pi - 3\pi r^2 = 0 \Rightarrow r^2 = \dfrac{161}{3} \Rightarrow r = 7.3257... \Rightarrow r \approx 7.33$ $\left[\dfrac{d^2V}{dr^2} =\right] - 6\pi r = [[-138.17, -138.09]]$ or -138 or -140 or awrt -44π < 0 hence maximum	M1 M1A1 M1A1 [5]
(c)	$V = \pi \times 7.3257\left(161 - 7.3257^2\right) = 2470$	M1A1 [2]
		Total 10 marks

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Part	Mark	Notes
(a)	B1	For finding a correct expression for h in terms of r This expression doesn't need to be simplified. Independent of method mark, so mark the expression for h in terms of r for this mark. For the final A mark, must be no errors.
	M1	For substituting their expression for h in terms of r into a correct expression for V $V =$ doesn't need to be present for this mark.
	A1* cso	For the correct formula exactly as written. $V =$ must be shown at least once, throughout working. Minimum steps as shown in the mark scheme, no errors or omissions. Steps shown in square brackets are not required, but are provided as if present, must be correct and must be checked by Examiners. All additional steps must also be checked.
ALT	B1	For finding a correct expression for rh Independent of method mark, so mark the expression for rh in terms of r for this mark. For the final A mark, must be no errors.
	M1	For substituting their expression for rh in terms of r into correct expression for V $V =$ doesn't need to be present for this mark.
	A1* cso	For the correct formula exactly as written. $V =$ must be shown at least once, throughout working. Minimum steps as shown in the mark scheme, no errors or omissions. Steps shown in square brackets are not required, but are provided as if present, must be correct and must be checked by Examiners. All additional steps must also be checked.
(b)	M1	For a minimally acceptable attempt (see general guidance, but also no power or r to increase) to differentiate the given expression. Simplification is not necessary.
	M1	Sets their derivative = 0 and **correctly** rearranges to finds a value for r or r^2 Allow 7.3 without a full rearrangement to imply this mark. This isn't a completely dependent mark, but must follow from anything considered to be an attempt to differentiate and then rearrange an equation with a 2 term expression in r Ignore any value of negative r
	A1	For the correct value of r. Accept awrt 7.33 Ignore any value of negative r Because candidates are told to use calculus, a correct answer will not imply method marks.
	M1	For a minimally acceptable attempt (see general guidance, but also no power or r to increase) to differentiate their expression for the first derivative which must have 2 terms, with a least one term in r
	A1	For a correct conclusion. The value of the second derivative doesn't have to be calculated, but if it is, the numbers in the interval, or 138 or 140 must be seen. Must state < 0 and a brief conclusion which could be as short as shown or # Don't award if there is work to consider a negative value of r, unless discounted or abandoned. **All work must be correct throughout (b) for this mark.** Must be a clear conclusion referring to only the positive value of r for this mark.
(c)	M1	For substituting their value of r into the given expression for V May be implied by a correct value.
	A1	awrt 2470 but not from 7.3 or incorrect r

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Question	Scheme	Marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 30 of 33 · Question 11 part c · 3 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

number		
11 (a)	$(600 =)2\pi r^2 + 2\pi rh$ oe eg $(300 =)\pi r^2 + \pi rh$	M1
	$h = \frac{300 - \pi r^2}{\pi r}$ oe | $\pi rh = 300 - \pi r^2$ oe	A1 cao
	$(V =)\pi r^2 "\left(\frac{300 - \pi r^2}{\pi r}\right)"$ oe | $V = "(300 - \pi r^2)"r$ oe	M1
	$V = 300r - \pi r^3$ *	A1* cso [4]
(b) (i)	$\frac{dV}{dr} = 300 - 3\pi r^2$	M1
	$0 = 300 - 3\pi r^2 \quad \rightarrow \quad r =$	M1
	$r = \sqrt{\frac{100}{\pi}}$ * cso	A1* cso
(ii)	$\frac{d^2V}{dr^2} = -6\pi r$	M1
	$\rightarrow \frac{d^2V}{dr^2} = -6\pi\sqrt{\frac{100}{\pi}}$ or $\frac{d^2V}{dr^2} = -6\pi \times 5.6418958.....$	
	When r is positive, $-6\pi r$ is negative (-106.347231…..) and therefore this value of r gives a maximum	A1 [5]
(c)	$(V =)300"\sqrt{\frac{100}{\pi}}" - \pi\left("\sqrt{\frac{100}{\pi}}"\right)^3 (= 1128.(379167))$	M1
	$p^3 = \frac{300"\sqrt{\frac{100}{\pi}}" - \pi\left("\sqrt{\frac{100}{\pi}}"\right)^3}{\frac{4}{3}\pi}$ (=269(.3806…))	dM1
	$p = 6.5$ cm	A1 [3]
		Total 12 marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 30 of 33 · Question 11 part c · 3 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	Correct expression for the surface area of the cylinder and an attempt to rearrange to $h =$ or $\pi rh =$ Allow errors in arithmetic but not mathematically incorrect process. πrh may be embedded, e.g. $300 = \pi r^2 + \pi rh$ becoming $300 = \pi r^2 + V$ would score M1A1M1 and may score full marks if correct final result obtained.
	A1	cao
	M1	Substitutes their expression for height or their expression for πrh into a correct expression for the volume.
	A1	cso no errors or omissions, must state $V =$
	Mark parts (i) and (ii) together.	
(b) (i)	M1	Minimally acceptable attempt at differentiation, see general guidance, no power to increase.
	M1	Places their derivative = 0 and attempts to rearrange to find r. Minimally acceptable derivative is of the form $a \pm b\pi r^2$
	A1	Correct value for r, exact value only. Must reject negative value if found, award A0 if not rejected.
(ii)	M1	Minimally acceptable attempt to differentiate their first derivative, see general guidance, no power to increase. Or testing gradients or a sketch.
	A1	Correct evaluation of second derivative or explanation of why second derivative is negative. Conclusion this value of r gives a maximum. No incorrect work.
(c)	M1	Correct substitution of their r into the expression for V.
	dM1	Attempts rearrangement using the formula for volume of a sphere to make p^3 the subject Correct order of operations applied to right hand side. Accept arithmetic slips.
	A1	$p = 6.5$ Accept awrt 6.5

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 30 of 33 · Question 11 part c · 3 marks

2022 JUN · Paper 02 (regional R) · extract 3 of 3 · find part c in this mark scheme extract

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Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 31 of 33 · Question 7 part c · 2 marks

2024 NOV · Paper 01 · find part c in this mark scheme extract

7 (a)	$\frac{1}{4}\pi r^2 + 2xr = 100 \quad \left(100 - \frac{1}{4}\pi r^2 = 2xr\right) \Rightarrow x = \frac{50}{r} - \frac{1}{8}\pi r$ oe for example $\frac{100 - \frac{\pi r^2}{4}}{2r}$ $\frac{400 - \pi r^2}{8r}$	M1
	$(P =)\frac{1}{4}(2\pi r) + 4x + 2r$ oe for example $(P =)\frac{1}{2}\pi r + 2x + 2x + 2r$	M1
	$(P =)\frac{1}{2}\pi r + 4\left(\frac{50}{r} - \frac{1}{8}\pi r\right) + 2r \left(= \frac{1}{2}\pi r + \frac{200}{r} - \frac{1}{2}\pi r + 2r\right)$ oe	M1
	$P = \frac{200}{r} + 2r$ *	A1*cso [4]
(b)	$\left(\frac{dP}{dr} =\right) -\frac{200}{r^2} + 2$	M1
	$0 = "-\frac{200}{r^2} + 2" \Rightarrow r =$	M1
	$(r =)10$	A1
	$\left(\frac{d^2P}{dr^2} =\right)\frac{400}{r^3}$	M1
	$r = 10, \frac{d^2P}{dr^2} > 0$ therefore minimum or $\frac{d^2P}{dr^2} = \frac{400}{10^3}$ or $\frac{2}{5} > 0$ therefore minimum	A1 [5]
(c)	$P = \frac{200}{("10")} + 2("10") = 40$	M1 A1 [2]
		Total 11 marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 32 of 33 · Question 8 part c · 2 marks

2022 JUN · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

8 a	Height of the waste paper basket $= \sqrt{(5x)^2 - (3x)^2} = 4x$ (cm)	M1
	$V = \frac{1}{2}(2x+8x)\times 4x \times h = 20x^2h = 2250 \Rightarrow h = \frac{2250}{20x^2}$ oe	M1
	$S = 2\times 20x^2 + 2xh + 2(5xh)$	M1
	$S = 40x^2 + 12x\left(\frac{2250}{20x^2}\right)$ oe	M1
	$S = 40x^2 + \frac{1350}{x}$ *	A1 cso (5)
b	$\frac{dS}{dx} = 80x - \frac{1350}{x^2}$ oe	M1
	$\frac{dS}{dx} = 80x - \frac{1350}{x^2} = 0$ so $x^3 = \frac{135}{8} \Rightarrow x = ...$	M1
	$x = 2.56$ awrt	A1
	$\frac{d^2S}{dx^2} = 80 + \frac{2700}{x^3} > 0$ for all positive values of x ∴ minimum	M1 A1ft (5)
c	When $x = 2.56$ $S = 40(2.56)^2 + \frac{1350}{2.56} = 789$ awrt	M1 A1 (2)
		Total 12 marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 32 of 33 · Question 8 part c · 2 marks

2022 JUN · Paper 02 · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	**M1**	For finding the height of the waste-paper basket using Pythagoras theorem. $\sqrt{(5x)^2-(3x)^2}=4x$ (cm)
	M1	For finding the volume which must come from; V = correct area of trapezium (using their height) × h $V=\frac{1}{2}(2x+8x)\times'4x'\times h=2250\Rightarrow('20'x^2h=2250)$ **and** for obtaining an expression for the height h: $h=\frac{2250}{'20'x^2}$ or $xh=\frac{2250}{'20'x}$ Please check their algebra carefully, as some may even substitute $hx^2=...$ **NB: This is an A mark in Epen.**
	M1	For writing the surface area in terms of 2 unknowns [x and h] which need not be simplified. This must be **correct** using their expression in terms of x for the height of the prism. e.g. $S=2\times\frac{(8x+2x)\times'4x'}{2}+2xh+2(5xh)=\left[40x^2+12xh\right]$
	M1	For eliminating h from their expression for S S must be in the form Ax^2+Bxh and h must be of the form $\frac{C}{x^2}$ where A, B and C are constants.
	A1 cso	For the given result exactly as written. $S=40x^2+\frac{1350}{x}$
(b)	**M1**	For an acceptable attempt to differentiate the **given** expression for S. See General Guidance.
	M1	For setting their $\frac{dS}{dx}=0$ and attempting to find a value for x The minimum acceptable expression is $\frac{dS}{dx}=Px\pm\frac{Q}{x^2}$ where P and Q are constants.
	A1	For awrt $x=2.56$
	M1	For differentiating their $\frac{dS}{dx}$, which must be as a minimum $\frac{dS}{dx}=Px\pm\frac{Q}{x^2}$ to find the second derivative to achieve as a minimum $\frac{d^2S}{dx^2}=\pm M\pm\frac{N}{x^3}$
	A1ft	Concludes that as $\frac{d^2S}{dx^2}$ will always be positive, [either by substitution, or by inference] so the value of x obtained will be a minimum. $\left[\frac{d^2S}{dx^2}=80+\frac{2700}{2.56^3}=240.9...\right]$ with a conclusion. **NOTE:** The ft only applies to their value of x. Do not ft an incorrect $\frac{d^2S}{dx^2}$
(c)	**M1**	Substitutes their value of x, obtained using a correct method) into the **given** expression for S [provided it is a positive value, do **NOT** allow negative values of x]
	A1	For awrt 789

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Answer 32 of 33 · Question 8 part c · 2 marks

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Question number	Scheme	Marks

Type MS · Optimisation part 2: finding, justifying and evaluating the maximum or minimum

Answer 33 of 33 · Question 7 part c · 2 marks

2022 JAN · Paper 01 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

7 (a)	(Surface area =) $12x^2 + 4xy + 3xy + 5xy = 144$ oe	M1
	$12x^2 + 12xy = 144$ oe leading to $\frac{144 - 12x^2}{12x}$ oe	A1
	$V = \frac{1}{2} \times 3x \times 4x \times y \ (= 6x^2 y)$	M1
	$= 6x^2 \times \frac{144 - 12x^2}{12x} = 6x(12 - x^2)$	M1
	$= 72x - 6x^3$ *	A1 cso [5]
(b)	$\left(\frac{dV}{dx}\right) = 72 - 18x^2$	M1
	$\left(\frac{dV}{dx} = \right)"72 - 18x^2" = 0$ $\Rightarrow x =$	M1 (A1 on ePen)
	$x = 2$	A1 (M1 on ePen)
	$\left(\frac{d^2V}{dx^2}\right) = -36x$	B1 (A1 on ePen)
	$x = 2 \ \frac{d^2V}{dx^2} < 0$ Therefore maximum	[4]
(c)	$V = 72 \times 2 - 6 \times 2^3 = 96$	M1 A1 [2]
		Total 11 marks

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Part	Mark	Notes
(a)	M1	For surface area as an equation in x and y (correct and unsimplified)
	A1	For $12x^2 + 12xy = 144$ oe and correctly making y the subject $\frac{144-12x^2}{12x}$ (oe)
	M1	$V = \frac{1}{2} \times 3x \times 4x \times y \ (= 6x^2 y)$
	M1	For correct substitution of their expression for y, which must be in terms of x only, into their formula for V.
	A1*cso	Obtains the given expression
(b)	M1	For attempting to differentiate V wrt x At least one term must be fully correct, the other – follow general guidance
	M1 (A1 on ePen)	For setting their derivative (which must involve a changed expression) = 0 and a complete solution to find the value of x
	A1 (M1 on ePen)	For $x = 2$ only
	B1 (A1 on ePen)	For a correct second derivative and correct justification that this is a maximum, with some form of conclusion. All work to be correct for this mark. i.e it is acceptable, as this is a simple substitution to state if $x = 2$, $\frac{d^2V}{dx^2} < 0$, but if a substitution of $x = 2$ is made, the value for the second derivative must be given as – 72.
ALT final B1	B1 (A1 on ePen)	Testing and substituting into the correct second derivative with appropriate values either side of $x = 2$. For correct justification that this is a maximum, with some form of conclusion. All work to be correct for this mark
(c)	M1	For correct substitution of their x into V
	A1	96

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Question number	Scheme	Marks